BTF Trace Viewer

Current version: **1.3.2** (Desktop Python app + Web app)

A PySide6-based interactive visualiser for FreeRTOS context-switch traces in **Best Trace Format** (`.btf`). Plain `.btf` files and compressed containers (`.btf.gz` / `.gz`, `.btf.bz2` / `.bz2`, `.btf.zip` / `.zip`) open the same way on Desktop and Web.

Screenshot

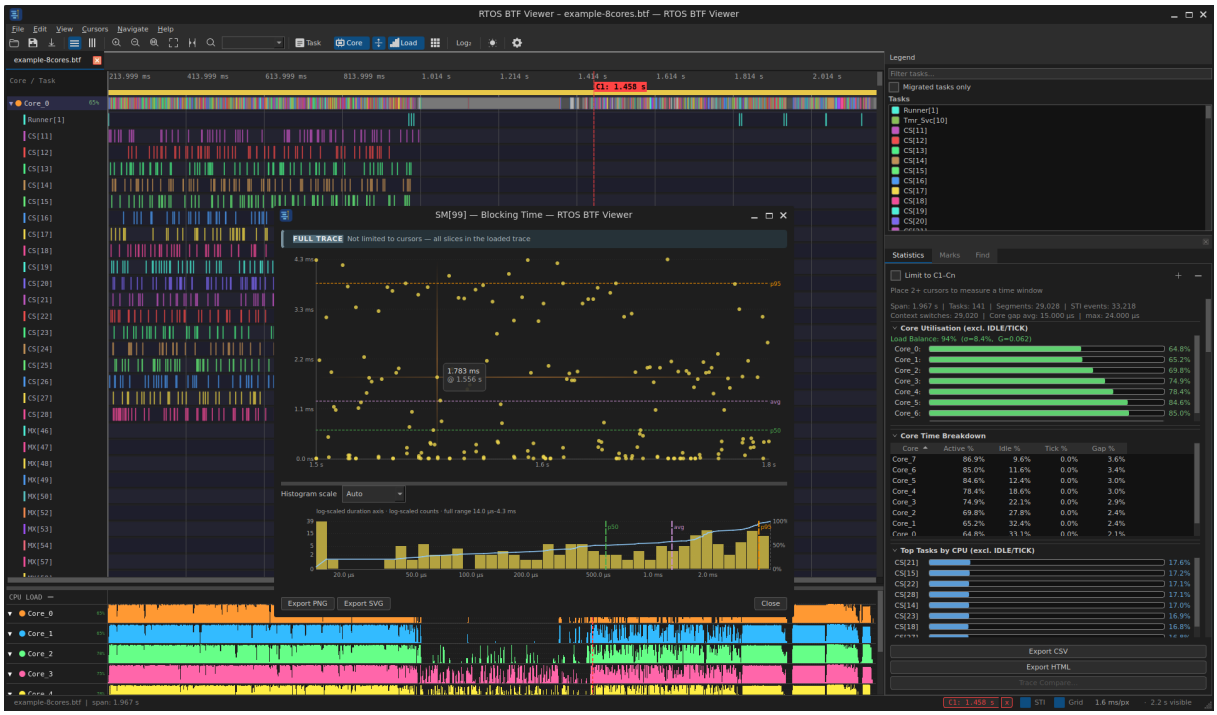


Figure 1: BTF Viewer screenshot

DEMO

Features

- **Two view modes** — Task View (one row/column per task) and Core View (one row/column per core, expandable)
- **Expand / Collapse all cores** — single-click toolbar button in Core View

- **Per-core expand / collapse** — click any core label to expand or collapse just that core
- **16-colour core palette** — up to 16 distinct core colours; cycles automatically beyond that
- **Deterministic colour mapping** — task and STI colours are assigned in a stable, repeatable sequence across runs
- **Horizontal and Vertical orientation** — switch at any time; active mode is highlighted in the toolbar; last used orientation is remembered across launches
- **Smooth zoom & pan** — mouse wheel, two-finger pinch (macOS), and keyboard shortcuts
- **Default zoom 2 timescale units/px** — the **1:1** toolbar button resets to 2 timescale units per pixel (for ns timescale, the UI shows 2 ns/px; configurable in Settings)
- **Zoom to cursor range** — Ctrl+R or the ⊡ **Range** toolbar button fits the viewport exactly between cursor C1 (left/top edge) and the last cursor (right/bottom edge)
- **Viewport culling** — only visible rows/columns and segments are rendered; no slowdown on large traces
- **Multi-tab traces** — open several .btf files at once (Desktop: closable tabs; Web: tab bar under the toolbar). Both restore open tabs, active tab, and per-tab zoom/cursors/marks/filters on launch (Desktop: btf_viewer.rc; Web: localStorage + IndexedDB trace cache)
- **Measurement cursors** — Desktop and Web support 2–8 cursors (default: 4); configurable in Settings
- **Trace compare** — with 2+ tabs open, **Trace Compare…** in the Statistics panel diffs **Summary**, **Top Tasks**, **Core Migrations**, **Blocking**, **Preemption**, and **Sync** side-by-side (Desktop + Web). Optional **Limit to each tab's cursor range** compares metrics within C1–Cn when 2+ cursors are placed on each trace
- **Core migration analysis** — detect tasks that run on multiple cores; **Core Migrations** stats table (ping-pong, STI correlation, gap-after vs other gaps), **Core-Pair Migration Summary** (per directed pair: count, bounces, avg gap), **Core Time Breakdown** (active/idle/tick/gap per core), **clickable migration heatmap** (pair×time for ≤ 16 cores; core×core matrix → outgoing pairs for larger traces → per-task sub-bins → timeline drill-down), **migration chord diagram** (directional core-to-core migration volume as a circular chord diagram; hover a core arc to highlight its migrations), **Migrated tasks only** legend filter, toolbar **All tasks** reset, **bounce-only filter** (Show: Bounce Only toggle

restricts the heatmap and chord diagram to lock-bounce migrations), and Find **Migrations** mode (Desktop + Web)

- **Core Affinity** — when traces include `affinity_set` STI events (from `vTaskCoreAffinitySet` / `traceENTER_vTaskCoreAffinitySet`), the **Core Affinity** statistics table shows each task's affinity mask history, observed execution cores, and flags violations in red — checked per slice against the mask in effect at that slice (before the first set the task is unrestricted) (Desktop + Web)
- **Analysis Findings** — toolbar **Analysis** opens a severity-tagged dialog (load imbalance, WCET/CPU hotspots, blocking, L/M/H priority inversion, core thrashing / hot pairs, deadline breaches, tick health, sync/mutex bounces) for the current Statistics scope; **Save as text…** writes a `.txt` copy. The same card is embedded in **Statistics → Export HTML** / headless `report --format html` (Desktop + Web)
- **Cursor-scoped statistics** — with 2+ cursors, the Statistics panel can limit all metrics (CPU%, execution slices, blocking time, inter-arrival, **preemption chain**, **priority inheritance**, **mutex / semaphore pairing**, **queue pairing**, **deadline violations**, scheduling summary, **Analysis Findings**, exports, and charts) to the window from C1 through the last cursor; toggle **Limit to cursor range (C1–Cn)** (Desktop + Web)
- **Cursor range summary** — with 2+ cursors, the status bar shows a quick min/max/avg segment summary (Desktop + Web); full per-task metrics remain in the **Statistics** panel
- **Task highlight** — hover or click any task label or Legend row to highlight all its segments; optional **Highlight segments on label hover** in Settings dims other tasks while hovering (off by default)
- **Dockable Legend panel** — colour swatches for every task, with a search box, **Migrated tasks only** filter, a **heatmap filter banner** (when drilled from the heatmap), and the same highlight interaction
- **Right-side panel** — **Statistics**, **Marks**, and **Find** tabs (Desktop: docked on the right, stacked below Legend; Web: tab bar on the right). Statistics holds metric tables; Marks holds cursors, bookmarks/annotations, and (on Web) Legend; Find searches tasks, annotations, migrations, STI events, interval spans, task lifecycle events, and object pointers
- **Tag View** — inspect tag channels/events (`tag_event`, `tag0_event` … `tag7_event`) alongside task/core activity
- **Metrics tables** — Execution Time Per Slice, **Blocking Time** (same metric as Tracealyzer **Response Time**: off-CPU gap / scheduling latency

between activations), **Inter-Arrival**, **Preemption Chain** (which tasks preempted whom), **Priority Inheritance**, **Mutex / Semaphore pairing** (with **Core Bounce** column flagging holds that crossed core boundaries), **Queue pairing** (send/recv by queue pointer), **Interval Analysis** (paired `interval_start` / `interval_stop` spans; when the BTF note includes `tid:{task_id}`, pairing is per interval id **and** task), **Tag Analysis** (per-channel min/avg/max/p95 for `tag0_event`…`tag7_event` samples), **Task Lifecycle** (per-task create/delete/suspend/resume event summary), **Core-Pair Migration Summary**, **Core Time Breakdown**, **Core Affinity** (affinity mask vs. observed cores), and **Deadlines / CPU budget** (per-slice deadline violations and global CPU budget threshold); click **Min** / **Max** (dotted underline) to jump and add an annotation at the BCET / WCET slice or shortest / longest gap (Desktop + Web)

- **Metrics distribution charts** — click any row in Execution Time, Blocking Time, Inter-Arrival, **Core Migrations** (in-dialog tabs to switch between **Dwell**, **Rate**, and **Gap**), Preemption Chain, **Priority Inheritance**, **Interval Analysis**, or **Tag Analysis** tables to open a scatter-plot + histogram popup; histograms use **adaptive scaling** (auto linear / p5–p95 / log duration, overflow buckets, optional log-scaled counts, CDF overlay); charts live-update when cursors move or cursor-range scope is toggled (Desktop + Web). On Desktop, each trace tab remembers its own open chart when you switch tabs
- **Segment tooltips** — hover any segment bar for duration, slice index on core, previous/next task on that core, and gap before the slice; time precision (decimal digits) is configurable in **Settings → Layout** and also applies to the hover-line badge, cursor labels, bookmarks/annotations, and the status bar
- **CPU Load Graph** — bar chart below the timeline showing per-core CPU utilisation; row labels show the **visible-window average** and, with 2+ cursors, a cursor-range average (`· C:xx%`); toggle with the **Load** toolbar button; drag the divider between timeline and CPU load to resize (Desktop + Web)
- **Resizable panels** — drag dividers between timeline and CPU load, the right-side panel (Statistics / Marks / Find) and Legend dock (Desktop), the **label column** (task names), and metric table sections in Statistics (Desktop + Web); splitter, label width, and table heights persist in `btf_viewer.rc` (Desktop) or browser `localStorage` (Web)
- **STI event markers** — software trace items rendered as coloured diamond markers
- **Find & Jump** — search tasks, annotations, or migrations; **Find** tab in the right panel (same as web); toolbar button and `Ctrl`+F; F3 / `Shift`+F3 steps through

matches

- **Bookmarks & Annotations** — mark important timestamps and attach free-text notes; persisted per trace file in `btf_viewer.rc`
- **Right-click context menu** — place/remove/clear cursors, add a bookmark, or add an annotation, all from a single right-click anywhere on the timeline
- **Recent files (Desktop)** — **File → Open Recent** lists the 8 most recently opened traces for one-click reopening
- **Dark / Light theme** — switch from **View → Switch to Light/Dark Theme** or **Settings → Appearance**
- **Export to PNG / SVG / clipboard** — capture the viewport in the **Snapshot Editor** (annotate with arrows, shapes, and text, then save or copy); direct clipboard copy also available from the menu
- **Export Perfetto** — **File → Export Perfetto…** (Desktop) or toolbar **Perfetto** (Web) writes Chrome Trace JSON openable in ui.perfetto.dev; also available via headless `perfetto` CLI
- **Headless image export (Desktop CLI)** — `snapshot` command renders the timeline, Migration Heatmap, or a statistics metric plot straight to PNG/SVG with no GUI, for scripting and CI (see Headless CLI)
- **Persistent settings** — all preferences stored in `btf_viewer.rc` alongside the script
- **Drag-and-drop** — drop a `.btf` (or `.gz` / `.bz2` / `.zip` containing a `.btf`) file directly onto the window
- **Optimised for large traces** — tested with up to **128 cores, 1 024 tasks, and 5 M+ events** (desktop). Web viewer: flat segment storage, parse worker, precomputed CPU-load bins, WASM-accelerated bisect/LOD, and debounced stats worker for 100k+ segment traces

Table of contents

Section	Contents
Statistics & metrics	Panel overview, Analysis Findings, metric tables, migrations, trace compare, charts
Find, marks & menus	Search, bookmarks, annotations, context menu, recent files
Session persistence	`btf_viewer.rc` and browser `localStorage`
Keyboard shortcuts	Desktop and web key bindings
Reference	`scripts/gen_trace.py`, BTF format, implementation map
WORKFLOWS.md	Practical diagnosis workflows (load balance, WCET, thrashing, exports, …)

Installation

Desktop viewer (`builds/btf_viewer.py`)

Requirements: Python 3.8+ and PySide6 ≥ 6.4.

```
# Install the only runtime dependency
pip install -r requirements.txt

# Run directly — no build step needed
python builds/btf_viewer.py [-h] [trace.btf]

# Headless CLI — see [Headless CLI (desktop
↪  only)](#headless-cli-desktop-only) in Desktop viewer
```

Developing the desktop viewer (source package) The desktop app is maintained as a multi-module package and merged into the single-file `builds/btf_viewer.py` for release (same model as the web viewer's committed `builds/btf_viewer.html`).

Mode	Command
Release / users	`python builds/btf_viewer.py [trace.btf]` — no build step
Dev (fast iteration)	From `BTFViewer/`: `python -m btf_viewer_pkg [trace.btf]`
Regenerate release builds	`make -C BTFViewer` → `builds/btf_viewer.html` + `builds/btf_viewer.py`
Desktop only	`make -C BTFViewer bundle`
Web only	`make -C BTFViewer web`
CI parity check	`make -C BTFViewer check-bundle` — fails if `builds/btf_viewer.py` drifts from `btf_viewer_pkg/`

Edit sources under `BTFViewer/btf_viewer_pkg/` (not the generated `builds/btf_viewer.py`), then run `make -C BTFViewer` and commit the package plus both files in `builds/`.

Makefile target	Action
`make` / `make all`	Web + desktop → `builds/btf_viewer.html` and `builds/btf_viewer.py`
`make web`	Web only → `builds/btf_viewer.html`
`make bundle`	Desktop only → `builds/btf_viewer.py`, `py_compile`, smoke `info` on `tracedata/example-2cores.btf.gz`
`make test`	Desktop characterization tests (`unittest`, offscreen Qt)

Makefile target	Action
`make check-bundle`	Same as `bundle`, then `git diff --exit-code builds/btf_viewer.py`

HiDPI / DPI scaling (Desktop) Qt 6 scales the desktop UI per monitor. Platform-specific tuning runs automatically at startup (`btf_viewer_pkg/platform.py` → `_configure_qt_startup()`).

Platform	Behaviour
Windows (native)	Uses Qt 6 per-monitor scaling. Stale `QT_FONT_DPI=96` (PyQt5 workaround) is cleared if set.
macOS (Retina)	Timeline/UI fonts use an ~11 px pixel baseline (`BTF_UI_FONT_PX`, default on macOS) so 8 pt settings match PyQt5 density on HiDPI.
WSLg (Wayland)	Reads Windows **AppliedDPI** (Control Panel → Display → Scale) via PowerShell and sets a **partial** `QT_WAYLAND_FORCE_DPI` nudge — not the full AppliedDPI, because WSLg already exposes matching `wl_output` scale and applying the full value doubles with native DPR (UI too large). Formula: `96 + (AppliedDPI − 96) / 4` (e.g. **120** at 200 %, **108** at 150 %).

Environment overrides (set before launch):

Variable	Purpose
`BTF_WSL_DPI=0`	Disable WSLg auto DPI nudge (use native WSLg / Qt scaling only).
`QT_WAYLAND_FORCE_DPI=N`	Force Qt Wayland logical DPI (e.g. 108, 120, 144). Overrides WSL auto-tuning.
`BTF_UI_FONT_PX=N`	Pixel baseline for UI and timeline monospace fonts (macOS default **11**). Use on any platform when pt sizing looks wrong.

Examples:

```
# WSLg — UI too small (try a larger nudge)
QT_WAYLAND_FORCE_DPI=144 python3 -m btf_viewer_pkg trace.btf

# WSLg — UI too large (disable auto-tuning or use a smaller nudge)
BTF_WSL_DPI=0 python3 -m btf_viewer_pkg trace.btf
QT_WAYLAND_FORCE_DPI=108 python3 -m btf_viewer_pkg trace.btf

# macOS / any platform — tweak font density without changing layout
BTF_UI_FONT_PX=13 python3 -m btf_viewer_pkg trace.btf
```

On WSLg, if scaling still does not match Windows, see microsoft/wslg#1335 for optional `.wslgconfig` compositor settings (`WESTON_RDP_HI_DPI_SCALING`, etc.).

Web viewer (`web/`)

A browser-based port of the viewer built with **Vue 3 + Vite**. It runs entirely in the browser — no server, no Python, and no backend required (use `make preview` or the hosted demo for best performance on large traces).

Requirements: Node.js 18+ and npm (build-time only).

Standalone single-file build (recommended) Produces a single self-contained `builds/btf_viewer.html` that can be opened directly in any browser — no server needed:

```
cd BTFViewer/web
# Demo trace embedded in the build (required for Demo button and fresh
↪  builds):
# web/example-2cores.btf.gz is committed; refresh from tracedata if needed:
#   cp ../../tracedata/example-2cores.btf.gz example-2cores.btf.gz
make          # installs deps and builds builds/btf_viewer.html
# or manually:
npm install
npm run build
```

Then open the build:

```
open BTFViewer/builds/btf_viewer.html   # macOS — double-click also works
```

The release artifact is `BTFViewer/builds/btf_viewer.html` (used by the hosted demo).

Do not open `BTFViewer/web/index.html` directly via `file://`; it is the Vite source entry used by the dev server.

`file://` vs local HTTP

Open method	Works?	Notes
Double-click `builds/btf_viewer.html`	Yes	Basic use; Chrome may block Web Workers on `file://`, so parsing falls back to the main thread (UI can freeze briefly on very large traces). Use **Open** to pick a `.btf` file.

Open method	Works?	Notes
`make preview`	**Recommended**	Serves the build over HTTP — Web Workers, WASM accel, and the stats worker all work as intended.
`make dev`	Dev	Hot reload at `http://localhost:5173`.
Hosted demo	**Recommended**	apps.kuoping.com/btf_viewer.html

For large traces (e.g. `tracedata/example-16cores.btf.gz` — 16 cores, 100k+ segments), prefer **`make preview`** or the hosted demo rather than `file://`.

Development server (with hot reload)

```
cd BTFViewer/web
make dev      # or: npm run dev
# → http://localhost:5173
```

Makefile targets

Target	Action
`make` / `make build`	Install deps + produce `builds/btf_viewer.html`
`make dev`	Start Vite dev server with hot reload
`make preview`	Serve the production build locally (recommended for large traces)
`make wasm`	Rebuild WASM timeline accelerator from `wasm/timeline_accel.wat`
`make clean`	Remove `builds/btf_viewer.html` and any stale `dist/`

Target	Action
`make dist-clean`	Same as `clean`, plus remove `node_modules/`
`make test`	Overlay draw smoke tests (`node --test tests/`)

macOS — install Node.js via Homebrew

```
brew install node
```

Windows / **Linux** Download the LTS installer from https://nodejs.org/ or use your system package manager (`winget install OpenJS.NodeJS`, `apt install nodejs npm`, etc.).

Desktop viewer

Interactive PySide6 GUI and optional **headless CLI** (same statistics engine as the Statistics panel).

Topic	Section
Launch & CLI	Usage · Headless CLI
HiDPI / DPI	HiDPI / DPI scaling (Desktop)
Timeline	View modes · Orientation · Zoom and pan
Analysis UI	Cursors · Legend · Export · Statistics & metrics · Settings

Usage

```
python builds/btf_viewer.py [-h] [trace.btf]
```

Passing a file on the command line opens that trace in a tab immediately. With no argument, the viewer restores the previous session from `btf_viewer.rc`.

Headless CLI (desktop only)

Command	Description
`info`	Trace summary on stdout (`--json` for machine-readable output)
`report`	Full statistics export (same as Statistics → Export CSV/HTML; HTML includes the Analysis Findings card)
`compare`	Two-trace diff (same as Trace Compare → Export)
`migrations`	Core Migrations table as CSV
`snapshot`	Export a PNG/SVG image — timeline, Migration Heatmap, or a statistics metric plot — without opening the GUI
`perfetto`	Export Chrome Trace JSON for ui.perfetto.dev (same as **File → Export Perfetto…**)

```
python builds/btf_viewer.py info trace.btf [--json] [--lo T] [--hi T]

python builds/btf_viewer.py report trace.btf -o|--output PATH [--format
↪  html|csv|both] [--lo T] [--hi T]

python builds/btf_viewer.py compare a.btf b.btf -o|--output PATH [--format
↪  html|csv|both] \
  [--name-a LABEL] [--name-b LABEL] \
  [--lo T --hi T | --lo-a T --hi-a T --lo-b T --hi-b T]

python builds/btf_viewer.py migrations trace.btf [-o PATH] [--lo T] [--hi T]
↪  # default: stdout
```

```
python builds/btf_viewer.py snapshot trace.btf -o|--output PATH --view
↪  timeline|heatmap|plot \
  [--format png|svg] [--task NAME] [--view-mode task|core] [--cpu-load] \
  [--metric
↪  tick|exec|block|inter|priority|preempt|mig_dwell|mig_rate|mig_gap] \
  [--preemptor NAME] [--lo T --hi T] [--drill-row N --drill-bin N] \
  [--width PX] [--height PX] [--theme dark|light]

python builds/btf_viewer.py perfetto trace.btf -o|--output PATH.json
```

The `snapshot --view`:

- `timeline` — the main task/core view (like **File → Save Image / Save SVG**); `--task` highlights and centers that task's row; omit `--lo`/`--hi` to **Fit to Window** (full span), or pass `--lo`/`--hi` to zoom a range. `--view-mode core` switches to **Core View** (default `task`); with `--task`, the CLI also filters to that task and expands every core it ran on so migrations appear as the same task hopping across core rows (see Highlight a migrating task). `--cpu-load` appends the synchronised **CPU Load** strip — with a locked `--task`, **Task View** shows that task's total utilisation; **Core View** shows **one sparkline per core** for that task (same as toolbar **Core** + **Load** + click-to-highlight).
- `heatmap` — the Migration Heatmap (core-pair × time-bin grid); `--lo`/`--hi` scope the grid; `--task` is not supported (the heatmap is inherently cross-task). `--drill-row`/`--drill-bin` (0-based, together) drill into the per-task grid for one core-pair row and time bin, matching a click in the GUI dialog — not supported for matrix-mode heatmaps (> 16 cores).
- `plot` — a statistics metric scatter + histogram popup, selected with `--metric`: `tick` (trace-wide tick-interval distribution, no `--task`), `exec` / `block` / `inter` / `priority` (require `--task`), `preempt` (requires `--task` **and** `--preemptor`), or `mig_dwell` / `mig_rate` / `mig_gap` (Core Migrations dwell/rate/gap distributions, require `--task`).

`--width`/`--height` only apply to `--view timeline` and `--view plot` (the heatmap image size is derived from its data grid). `--task` accepts the display name (e.g. `Producer[1]`), the bare name without `[id]`, or the raw merge key, matched case-insensitively.

Files can also be opened via **File → Open** (`Ctrl+O`), **File → Close Tab** (`Ctrl+W`), **Ctrl+Tab** / **Ctrl+Shift+Tab** (cycle tabs), or drag-and-drop onto the window.

View Modes

Mode	Description
Task View	One row (horizontal) or column (vertical) per task across all cores; core tint applied to segment bars
Core View	One expandable row/column per CPU core; bars coloured by running task

In **Core View**:

- Click a core label to **expand** or **collapse** that individual core's per-task sub-rows/columns.
- Use the ⊞ **Expand All** / ⊟ **Collapse All** toolbar button to expand or collapse every core at once.
- Works in both **Horizontal** and **Vertical** orientations.

Orientation

- **Horizontal** (default) — time runs left to right; task/core labels are on the left
- **Vertical** — time runs top to bottom; task/core labels are at the top

Switch orientation using the ↔ **Horizontal** / ↕ **Vertical** toolbar buttons or **View → Horizontal layout / Vertical layout**. The active orientation button is highlighted.

In **Horizontal** mode (default), drag the visible splitter on the **right edge** of the task-name label column to resize it (60–600 px). Double-click that edge to auto-fit the widest visible label. Width is saved to `btf_viewer.rc` (`[view] label_width` + per-window profile) on Desktop, or `labelWidth` in `btf-viewer-settings-v1` on Web. In **Vertical** mode, double-click the bottom edge of the label row for the same auto-fit (Desktop + Web).

In **Vertical** mode: - The ruler column (left edge) is frozen and always shows time labels as you scroll horizontally. - The label row (top edge) is frozen and always

shows task/core names as you scroll vertically. - The top-left corner area shows the **TICK** band label when TICK events are present. - Drag the bottom edge of the label row to resize label height (Desktop); persisted under the same `label_width` / `labelWidth` keys as horizontal mode.

Task Labels

Regular task labels show the task name and task ID, for example `MyTask[3]` or `Worker[0x8]`. Raw BTF task entities may use `[core/id]name`, `name[id]`, or `name(0x…)` / `name(dec)` — see Task label naming. IDLE and TICK tasks show their bare name (`IDLE`, `IDLE0`, `IDLE1`, etc.) without an ID suffix. IDLE tasks always render in grey; each IDLE task on a different core gets a distinct shade.

Task Highlight

Hovering or clicking a task name in the label column or Legend panel highlights all timeline segments for that task.

Action	Effect
Hover over a task label or Legend row	Transiently highlights that task’s segments
Hover leave	Removes the transient highlight and restores any persistent highlight
Click a task label or Legend row	Locks the highlight on that task persistently
Click the same locked task again	Cancels the persistent highlight
Click empty area in the label column	Cancels the persistent highlight
Click empty area in the Legend panel	Cancels the persistent highlight

When **Highlight segments on label hover** is enabled in Settings (off by default), hovering a label also **dims** segments of other tasks. Leave it off on very large traces for smoother scrolling.

When a task is persistently highlighted, its row gets a colour tint, its label turns gold and bold, and its segment bars show a white border. Hovering another task while a lock is active shows both highlights at the same time.

Cursors

Between 2 and 8 cursors can be placed on the timeline (default: 4; adjustable in **Settings → Layout → Max cursors**). Delta times between consecutive cursors are shown on the timeline and in the status bar.

Placing and Moving

Action	Effect
Left-click on the timeline area	Place a cursor, or **remove** one if the click is near an existing cursor line
Shift+left-click	Snap the new cursor to the nearest segment boundary within 8 px (Desktop + Web)
Drag a cursor line	Move it to a new time position
`C` (keyboard)	Place a cursor at the viewport centre

Removing

Action	Effect
Right-click → **Remove nearest cursor**	Remove the cursor closest to the click position
Right-click → **Clear all cursors**	Remove all cursors at once
`Shift+C`	Clear all cursors

Action	Effect
Drag a status-bar cursor badge out of the status bar	Remove that specific cursor

Navigating

Action	Effect
Click a `C1` / `C2` / … badge in the status bar	Scroll the view to that cursor
`Ctrl+R` / ⊡ **Range** toolbar button	Zoom view to fit exactly between C1 and the last cursor

Cursor range summary (status bar)

When two or more cursors are placed, the **status bar** shows a quick summary of segment durations for slices that start **and** end within the cursor range: **min**, **max**, and **avg**.

For full per-task/per-core metrics scoped to the cursor window, use the **Statistics** panel — see Statistics & metrics and **Limit to cursor range (C1–Cn)**.

Multi-tab traces (Desktop)

Action	Effect
File → Open (`Ctrl+O`)	Open a trace in a **new tab** (or switch to it if already open)
File → Close Tab (`Ctrl+W`)	Close the active tab

Action	Effect
Ctrl+Tab / **Ctrl+Shift+Tab**	Next / previous trace tab
Click a tab	Switch the timeline, legend, statistics, marks, and CPU load graph to that trace
× on a tab	Close that tab

Each tab has its own cursors, zoom level, bookmarks, annotations, find state, and metrics chart session. Shared settings (theme, orientation, row height, etc.) apply to all tabs.

On exit, the viewer saves the list of open tab paths, the active tab index, and per-tab zoom/cursor layout to `btf_viewer.rc`. The next launch reopens the same tabs automatically (unless a file path on the command line overrides session restore).

Legend Panel

The Legend lists every task with its colour swatch and `Name[id]` label.

- The panel is a dockable window; it can be detached, closed, and re-opened via its × button.
- Toggle visibility from **Settings → Display → Legend panel** (`Ctrl+,`).
- A **Search** box at the top filters the displayed task list.
- **Migrated tasks only** (Web: checkbox in Legend on the Marks page; Desktop: same filter in the legend dock) hides tasks that ran on a single core only — useful with core migration analysis.
- After a **heatmap drill-down**, a blue **Heatmap: … (N)** banner appears with a **Clear** button; it filters the legend list to match the timeline.
- Hover and click Legend rows to highlight tasks using the same rules as the label column.

Zoom and Pan

Wheel and trackpad gestures depend on **orientation** (see Orientation). `Shift` swaps the time-axis and row/column-axis mappings.

Horizontal orientation (time left → right)

Action	Effect
`Ctrl` + Scroll wheel	Zoom in or out centred on the pointer
Two-finger pinch (macOS)	Zoom in or out
Vertical trackpad swipe / vertical scroll wheel	Scroll task/core rows vertically
Horizontal trackpad swipe	Pan along the time axis
`Shift` + scroll wheel	Swap axes (vertical swipe pans time, horizontal swipe scrolls rows)
`Ctrl+0`	Fit entire trace to window
1:1 toolbar button	Reset to default zoom (2 timescale units/pixel; for `ns` timescale, UI shows 2 `ns/px`; configurable in Settings)
Toolbar zoom+ / zoom− buttons	Zoom in or out by 2×

Vertical orientation (time top → bottom)

Action	Effect
`Ctrl` + Scroll wheel	Zoom in or out centred on the pointer
Two-finger pinch (macOS)	Zoom in or out
Vertical trackpad swipe / vertical scroll wheel	Pan along the time axis
Horizontal trackpad swipe	Scroll task/core columns horizontally

Action	Effect
`Shift` + scroll wheel	Swap axes (vertical swipe scrolls columns, horizontal swipe pans time)
`Ctrl+0`	Fit entire trace to window
1:1 toolbar button	Reset to default zoom (2 timescale units/pixel; for `ns` timescale, UI shows 2 `ns/px`; configurable in Settings)
Toolbar zoom+ / zoom− buttons	Zoom in or out by 2×

Export

Snapshot Editor (PNG)

File → Save as Image (PNG)… (`Ctrl+S`), the toolbar **Save PNG** / **Shot** buttons, and the plot-dialog **Export PNG** action all capture the current viewport and open the **Snapshot Editor** — they do **not** write a file immediately.

In the editor you can draw annotation shapes (arrow, double arrow, line, rectangle, circle, text) and then:

- Check **Dash** to draw dashed strokes (lines, arrows, rectangles, circles).
- Click a shape to select and drag it; **Ctrl+click** duplicates the shape (then drag to position the copy).
- **Double-click** any shape to add or edit its label inline (lines/arrows: centered, aligned to the segment; rectangles/circles: centered).
- Single-click with the **Text** tool adds a standalone text label inline.
- **Save PNG…** — write the annotated image to disk (includes CPU load graph when **Load** is on).
- **Copy to Clipboard** — copy the annotated image.

Direct clipboard copy

File → Copy Image to Clipboard (`Ctrl+Shift+C`) copies the raw viewport (timeline + CPU load when visible) without opening the editor. On Linux, `xclip`, `xsel`, or `wl-copy` is used when available (Qt clipboard is unreliable for images on X11/Wayland).

SVG

File → Save as SVG… (`Ctrl+Shift+S`) or the toolbar **Save SVG** button exports the current viewport as vector SVG (includes CPU load when visible).

Perfetto (Chrome Trace JSON)

File → Export Perfetto… (`Ctrl+Shift+E`, Desktop) or the toolbar **Perfetto** button (`Ctrl+Shift+E`, Web) writes a Chrome Trace Event Format `.json` file for the loaded trace. Open it in ui.perfetto.dev (**Open trace file**).

The export includes:

Process	Tracks
Cores	One row per core; duration slices named by the running task
Tasks	One row per task; on-CPU run slices (core in args); migration markers
STI	Instant events per software-trace channel, plus TICK marks
Intervals	Paired `interval_start` / `interval_stop` spans (when present)

Headless (Desktop): `python3 builds/btf_viewer.py perfetto trace.btf -o trace.json`

Analysis Findings

Toolbar **Analysis** (Desktop + Web) opens a dialog of heuristic findings for the current Statistics scope (full trace, or **Limit to cursor range** when enabled). Severity is `info` / `warning` / `error`; lists are capped (top ~5). **Save as text…** writes a plain-text `.txt` copy.

Theme	Typical trigger
Load imbalance	Score < 70 % or σ > 30 % (warn); Score ≥ 85 % and σ ≤ 30 % → “reasonably balanced”
WCET / CPU hotspots	Highest CPU% tasks
Blocking	Tasks with many off-CPU gaps
Priority inversion	L/M/H pattern in Priority Inheritance
Thrashing / hot pairs	High Migr Rate, Ping, short Dwell; high Bounce % pairs
Deadlines / CPU budget	Configured thresholds breached
Tick health	Status not GOOD, or missed-tick estimate > 0
Sync / mutex bounces	Core bounce > 0 or pairing issues

Findings are **not** a Statistics panel section. The same card is embedded near the top of **Statistics → Export HTML** and headless `report … --format html`. Practical walkthroughs: WORKFLOWS.md.

Settings

Open **Settings** from the toolbar (⚙ **Settings**) or via **View → ⚙ Settings…** (`Ctrl+,`).

	Desktop	Web
Storage	`btf_viewer.rc` in the viewer directory	Browser `localStorage` key `btf-viewer-settings-v1`
When saved	Immediately on each change	Immediately on **Save** (or when closing with **Save**); **Cancel** reverts unsaved edits
Live preview	—	Changes apply to the open trace while the dialog is open; cancel restores the pre-open snapshot

Preferences are restored on the next launch (desktop) or page reload (web).

Appearance

Setting	Description
Theme	**Dark** (default) or **Light**; toggle with the `D` shortcut
Timeline labels	Font size for task/core labels drawn on the timeline (pt on Desktop, px on Web; default **8**)
UI / menus	Font size for menus, toolbar, and status bar (pt on Desktop, px on Web; default **8**)
Colorblind-safe palette	Switches the task colour palette to the **Okabe-Ito 8-colour** set, designed to be distinguishable under the most common types of colour-vision deficiency (deuteranopia, protanopia, tritanopia). Off by default.

On Desktop, macOS Retina uses pixel-based font sizing by default; override with `BTF_UI_F0NT_PX` (see HiDPI / DPI scaling).

Display

Setting	Description
Legend panel	Show or hide the dockable Legend panel
Statistics panel	Show or hide the dockable Statistics panel
STI events	Show or hide software-trace item marker rows
Grid lines	Overlay vertical grid lines on the timeline (**on** by default)
Highlight on label hover	Dim all other segments when hovering a task label (**off** by default; enable for emphasis, disable for better performance on large traces)

Layout

Setting	Description
Label column	Width of the frozen task/core label column (60–600 px). Drag the timeline splitter (Desktop + Web) or set here; persisted in `btf_viewer.rc` / `btf-viewer-settings-v1`
Row height	Height of each task/core row (12–60 px; default **22**)
Row gap	Vertical gap between rows (0–20 px; default **4**)
1:1 zoom level	Target zoom of the **1:1** button and the maximum zoom-in limit (0.5–200 timescale units/px; UI unit follows trace timescale, e.g. `ns/px`; default **2**)

Setting	Description
Max cursors	Maximum number of simultaneously visible cursors (4–8; default **4**)
Time display precision	Decimal digits shown in tooltips, cursor labels, the hover-line badge, bookmarks/annotations, and the status bar (0–9; default **3**)
CPU load row height	Height of each CPU load graph row (16–120 px; default **30**)
STI row height	Collapsed STI channel row height (12–60 px; default **18**)
STI waveform height	Expanded tag-event waveform row height (40–300 px; default **80**)
STI waveform style	Y-axis scale for expanded tag waveform rows: **Linear** (default) or **Log₂**. Toggle with the **Log₂** toolbar icon (Web: icon-only; hover for the label).

Analysis

Available under **Settings → Display → Analysis thresholds** on both Desktop and Web.

Setting	Description
CPU budget %	Global CPU budget threshold (0–100 %, step 0.1 %; **0 = off**). Any task whose CPU% in the current scope exceeds this value appears in the **CPU budget exceeded** violation table in the Statistics panel.

Setting	Description
Task deadlines	Per-task deadline thresholds in nanoseconds; one entry per line in the form `TaskName=nanoseconds` (e.g. `Worker[0]=500000`). Task names must match the display name shown in the label column. Any execution slice longer than the configured threshold is listed in the **Slice over deadline** violation table.

Changes take effect immediately when the Settings dialog is accepted. The **Deadlines / CPU budget** section of the Statistics panel shows a prompt to open Settings when no thresholds are configured.

Default values (Desktop + Web)

These match the compiled defaults in `btf_viewer.py` (`USER CONFIGURATION`) and `web/src/utils/settingsStore.js` (`DEFAULT_SETTINGS`). Desktop stores font sizes in **pt**; the web viewer uses the same numeric values in **px**.

Setting	Default
Theme	Dark
Timeline label font	8
UI / menus font	8
Show Legend / Statistics / Marks / STI / CPU load	On
Grid lines	**On**
Highlight segments on label hover	**Off**
Colorblind-safe palette	Off
Label column width	160 px
Row height	22 px
Row gap	4 px

Setting	Default
STI row height	18 px
STI waveform height	80 px
STI waveform style	Linear
1:1 zoom level	2 timescale units/px
Max cursors	4
Time display precision	3 decimal digits
CPU load row height	30 px

Existing saved settings (`btf_viewer.rc` or browser `localStorage`) are not overwritten when defaults change — use **Reset to defaults** in the Settings dialog (or clear `btf-viewer-settings-v1`) to pick up new defaults.

Web viewer

Browser-based viewer (Vue 3 + Vite). Feature parity with the desktop app for timeline, statistics, and trace compare; rendering uses Web Workers, PixiJS, and optional WASM acceleration.

Toolbar: icon-only controls (hover for tooltips) so the bar fits narrow windows; overflow ⋯ holds controls that still do not fit.

No CLI — scripted export uses desktop `btf_viewer.py` (`report`, `compare`, `migrations`, `snapshot`, …).

Topic	Section
UI & files	Opening a file · View modes · Multi-tab traces
Layout	Right panel · CPU load · Session restore
Performance	Web performance architecture

View modes

Same two modes as the desktop viewer:

Mode	Description
Task View	One row per task, coloured by task identity
Core View	One row per CPU core; bars coloured by running task. Click the ▶ arrow in the label column to expand a core into per-task sub-rows

Toggle with the **Task** / **Core** toolbar icons (hover for the label).

Opening a file

Click the **Open** toolbar icon and select any `.btf` file (or a compressed container: `.btf.gz` / `.gz`, `.btf.bz2` / `.bz2`, `.btf.zip` / `.zip`). Each file opens in a new **tab** in the bar below the toolbar; click a tab to switch traces, or **×** to close one. Opening the same filename again focuses the existing tab. On the Web build, the **Demo** icon loads the embedded `example-2cores.btf.gz`.

Parsing: traces are parsed in a **Web Worker** when the page is served over HTTP (`make dev`, `make preview`, or the hosted demo). On `file://` URLs some browsers block workers, so parsing may run on the main thread instead. A progress overlay (`Reading…` / `Parsing…` / `Opening trace…`) is shown until the timeline is ready.

Large traces: the web viewer is optimised for high segment counts (flat segment storage, precomputed CPU-load bins, WASM-accelerated bisect/LOD, debounced stats worker). On first paint after load it briefly uses a coarse LOD (like pan/zoom) and upgrades to full quality within a few hundred milliseconds — this keeps the UI responsive on traces such as `tracedata/example-16cores.btf.gz` (16 cores, 100k+ segments).

Sample traces in the repo: `tracedata/example-2cores.btf.gz` (small 2-core demo, also embedded in the web build), `tracedata/example-4cores.btf.gz` (4-core SMP, good for statistics demos), and `tracedata/example-16cores.btf.gz` (large SMP).

Zoom & pan

Wheel and trackpad gestures depend on **orientation** (↔ `Horizontal` / ↕ `Vertical` toolbar buttons). `Shift` swaps the time-axis and row/column-axis mappings. See Zoom and Pan for the full table (Desktop and Web use the same rules).

Horizontal orientation (default): vertical swipe scrolls rows; horizontal swipe pans time.

Vertical orientation: vertical swipe pans time; horizontal swipe scrolls columns.

Action	Effect
`Ctrl` + scroll wheel	Zoom in / out centred on mouse pointer
Vertical trackpad swipe / vertical scroll wheel	Scroll rows (horizontal layout) or pan time (vertical layout)
Horizontal trackpad swipe	Pan time (horizontal layout) or scroll columns (vertical layout)
`Shift` + scroll wheel	Swap the two axes above
+ / **−** toolbar buttons	Zoom in / out around viewport centre
Fit toolbar button	Fit the entire trace into the viewport

Cursors

Between **2 and 8** cursors can be placed (default: **4**; adjustable in **Settings → Layout → Max cursors**). Delta times between consecutive cursors are shown in the **Marks** tab (Web) or status bar (Desktop).

Action	Effect
Left-click on timeline	Place a cursor
Left-click near an existing cursor	Remove it
× **Cursors** toolbar button	Clear all cursors

Task highlight

Hover any task label (left column) or **Legend** swatch to highlight all segments for that task. Click to lock the highlight; click again to release.

By default, hovering does **not** dim other tasks — enable **Settings → Display → Highlight segments on label hover** for that behaviour (better performance on large traces when left off).

Grid lines & dark/light theme

Grid lines are **on** by default (vertical time guides). Toggle with the **grid** toolbar button or **Settings → Display → Grid lines**.

Toggle dark/light theme with the **moon** toolbar button or **Settings → Appearance**. The default theme is **dark**.

CPU Load Graph

A bar chart below the timeline shows per-core (or total) CPU utilisation over the currently visible time window.

- **Toggle** — click the **Load** toolbar button to show or hide the graph.
- **View modes** — in **Task View** a single *CPU Load* row shows aggregate utilisation; in **Core View** each core gets its own row.
- **Task highlight** — lock-highlight a task (click its label): **Task View** switches the strip to **that task's** total utilisation; **Core View** shows **one row per core** with that task's usage on each core. Clear the highlight to return to aggregate / all-tasks load.
- **Fit + Core View + highlight workflow** — `Ctrl+0` (Fit to Window) → toolbar **Core** → click a task → enable **Load**. Headless:

```
python builds/btf_viewer.py snapshot ../tracedata/example-8cores.btf.gz \
    -o ../images/stats/tasks-cpu-load-cs22.svg \
    --view timeline --view-mode core --task "CS[22]" --cpu-load --height 900
```

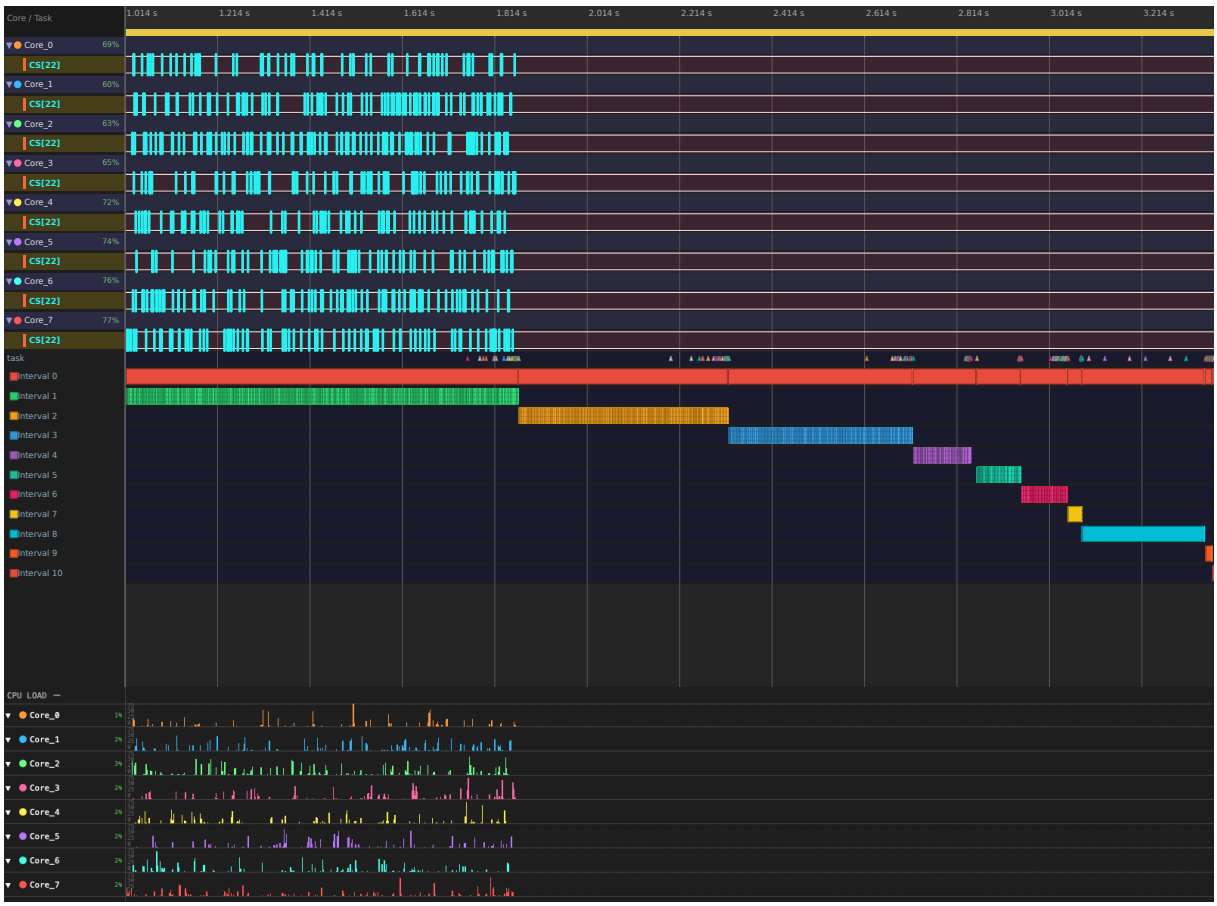


Figure 2: Core View: CS[22] highlighted, Fit to Window, per-core CPU Load for that task

CS[22] locked in Core View (Fit to Window). Timeline: task hops across Core_0...Core_7. ***CPU Load*** *strip: one sparkline per core for* ***that task only*** *(~1–3 % each while it runs in the CS phase; flat after ~1.8 s).*

- **Row labels** — each row shows average load over the **currently visible** time window; with 2+ cursors placed, labels also show the average over the cursor range (· C:xx%), and the graph shades the C1–Cn window in blue.
- **Expand / Collapse** — in Core View click a core row header to collapse it to a compact bar.
- **Hover** — moving the pointer over the timeline projects a live cursor onto the load graph; each row shows a load % badge at the hover time (no timestamp on the graph).

- **Resize** — drag the horizontal bar between the timeline and CPU load panel to change the CPU load height.
- **Cursors, bookmarks & annotations** appear as vertical lines in the load graph (no text labels on the graph itself).

Right panel & layout

The right side uses three tabs — **Statistics**, **Marks**, and **Find** — with a tab bar at the bottom of the panel (same on Desktop and Web).

Tab	Contents
Statistics	Collapsible metric tables, exports, **Trace Compare…**
Marks	**Web:** Cursors, cursor-range summary, unified bookmarks/annotations list, Legend. **Desktop: Curs.** / **Bookm.** / **Anno.** sub-tabs, cursor-range label, Legend in a separate dock above this panel
Find	Search (Contains / Exact / Regex / Migrations); `Ctrl`+F focuses this tab

- **Resize** — drag the vertical bar between the timeline and the right panel to change panel width.
- **Legend (Desktop)** — separate dock above the Statistics/Marks/Find panel; toggle via **Settings → Display → Legend panel**.
- **Legend (Web)** — section inside the **Marks** tab when **Legend panel** is enabled in Settings.
- **Migration heatmap** — toolbar **Heatmap** button (multi-core traces only). ≤ 16 cores: pair grid → task grid → timeline zoom/filter. > 16 cores: core×core matrix (row click) → outgoing pairs → tasks. **Export PNG / SVG** of the current drill level from the heatmap dialog. Toolbar **All tasks** appears while filtered. See Migration heatmap.
- **Migration chord diagram** — toolbar **Chord** button (multi-core traces only). Circular diagram with one arc per core; hover an arc to highlight its migrations and dim the rest. **Show: All Migrations / Show: Bounce Only** toggle restricts it to lock-bounce migrations. **Export PNG / SVG** of the current view. See Migration chord diagram.

- **Analysis Findings** — toolbar **Analysis** button opens heuristic findings for the current Statistics scope; **Save as text…** exports a `.txt` copy. See Analysis Findings.

Multi-tab traces (Web)

Same tab bar behaviour as desktop: each `.btf` opens in its own tab with independent cursors, marks, zoom, chart state, Find queries, and undo history. Use **Ctrl+Tab** / **Ctrl+Shift+Tab** to cycle tabs. Switching tabs closes an open **Migration heatmap** or **Migration chord diagram** dialog. **Trace Compare…** uses any two loaded tabs — useful for before/after or build-to-build diffs.

Session restore (Web)

The web viewer persists **settings** in browser `localStorage` (key `btf-viewer-settings-v1`: font sizes, label column width, row height, max cursors, theme, etc.) and **session state** in `btf-viewer-session-v1` (view mode, orientation, panel widths, stats table heights, open tab names, active tab, and per-tab viewport/cursors/marks/filters). Parsed trace payloads for recently opened files are cached in **IndexedDB** (`btf-viewer-traces-v1`, up to 8 traces) so tabs can reopen after refresh without re-selecting files. Settings are saved when you accept the Settings dialog or finish a label-column drag; session state is debounced (~400 ms) when you change tabs, cursors, marks, viewport, filters, or panel sizes.

On page load:

- Restores global settings from `btf-viewer-settings-v1` (including **label column width** 60–600 px).
- Restores global view options from `btf-viewer-session-v1` (task/core mode, orientation, grid, STI, CPU load, dark mode).
- Restores right-panel width, CPU-load panel height, and stats table section heights.
- Reopens cached trace tabs (when IndexedDB still holds the file data), restores the last active tab, and reapplies each tab’s zoom, cursors, marks, and legend filters.

Not persisted: trace files you never opened in this browser, or traces evicted from the IndexedDB cache (LRU cap). Re-open those with **Open** or **Demo**.

Limitations:

- `localStorage` / IndexedDB may be unavailable in strict private browsing modes.
- SVG timeline export includes cursor Δ badges between time-sorted cursors (same as on-screen ruler).

Web performance architecture

The web viewer shares the desktop feature set but uses a different rendering pipeline tuned for the browser:

Stage	Implementation
Parse	`btfParser.js` in a Web Worker; results packed via `tracePack.js` (flat `SegStore`, index arrays per task/core/LOD tier)
Transfer	Structured clone to the main thread (no transferable buffer detach issues)
CPU load	Bins precomputed at parse time (`cpuLoadBins.js`) — the CPU Load panel reads bins, not raw segments
Timeline paint	Hybrid **PixiJS WebGL** (segment fills, batched by color) + **Canvas 2D** chrome (ruler, STI, outlines, labels); viewport culling, LOD binning, GPU segment budget up to ~120k/frame, optional WASM bisect (`wasmAccel.js`); falls back to Canvas 2D-only if WebGL is unavailable
Parse / LOD (large traces)	Web Worker parse; WASM accelerates LOD bin de-duplication (`lod_summary_indices`) and bulk start-array gather (`gather_starts`) during pack; worker skips main-thread yield delays

Stage	Implementation
Statistics tables	Summary metrics on the main thread; expanded execution/blocking/inter-arrival tables in a debounced stats worker (`statsWorker.js`); preemption chain on the main thread
Initial load	Coarse “load-settle” paint, then full quality; WASM upload deferred to idle time so the first frame is not blocked

Chrome DevTools may log `[Violation] 'requestAnimationFrame' handler took …ms` on very large traces during the final full-quality repaint — that is a performance hint, not an error.

Statistics panel — cursor-scoped metrics

When **2 or more cursors** are placed, check **Limit to cursor range (C1–Cn)** at the top of the Statistics panel (enabled by default). All summary counts, CPU tables, execution/blocking/inter-arrival metrics, Analysis Findings, CSV/HTML export, and distribution charts then use only data inside the cursor window:

Metric	Scoping rule
Span / summary counts	Range width; tasks/segments/STI with any overlap
Core & task CPU %	Overlapping active time ÷ range width
Execution time per slice	Only slices **fully inside** the range
Blocking time	Off-CPU gap between consecutive slices; only pairs where **both** slices are fully inside the range
Inter-arrival	Activations whose start time falls inside the range
Preemption chain	Preemption overlaps counted only when the victim’s blocking gap and the preemptor overlap are inside the range
Interval analysis	Paired spans whose start/stop times overlap the range

Metric	Scoping rule
Core migrations	Migration events and per-core active time with overlap in the range

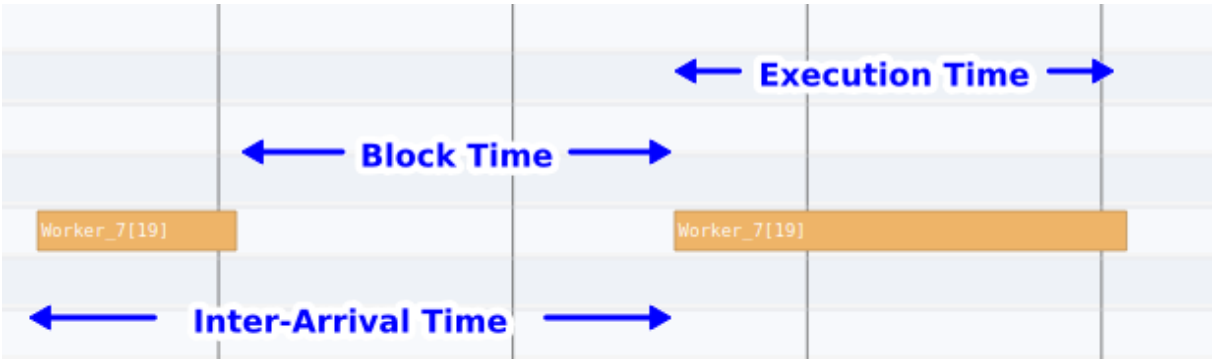


Figure 3: Statistics panel with cursor-scoped metrics

Uncheck the box to return to full-trace statistics.

Below the scope checkbox, a **scheduling summary** line shows context-switch count and average/max core gap (idle time between consecutive slices on each core). Metric tables are **collapsible** — click a section title to expand or collapse it. Drag the handle below a metric table to resize its height.

Section	What it shows
Core Utilisation	Active (non-IDLE, non-TICK) CPU time per core as a percentage; **Load Balance Score** gauge (green / amber / red zones)
Top Tasks by CPU	Top 10 worker tasks ranked by total CPU time
Trace Health (TICK)	STI TICK period regularity, **tick / tickless mode detection**, large gaps, missed-tick estimate, and **Tick Distribution** chart (Desktop + Web: whenever ≥ 2 ticks in scope)
Core Migrations	Per-task cross-core migration stats (see Core migration analysis)

Section	What it shows
Execution Time Per Slice	Per-task slice duration stats (runs, CPU%, min/avg/max/p95)
Blocking Time	Off-CPU gap between consecutive activations of the same task (Tracealyzer **Response Time** — identical definition)
Inter-Arrival Time	Gap between successive activation start times
Preemption Chain Analysis	For each victim task, which preemptors ran during its off-CPU gaps
Priority Inheritance	When traces include `create pri:N` and priority STI events (`priority_inherit` / `priority_disinherit` / `set_priority`): tasks boosted above base priority
Mutex / Semaphore pairing	Pairs `take`/`give` STI events by object pointer (`0x........`); flags orphan gives, cross-task gives, unmatched takes, delete-while-held, **core-boundary lock bounces (`CORE_MIGRATION_WHILE_HELD`)**, and multi-mutex hold at trace end
Interval Analysis	Paired `interval_start` / `interval_stop` spans per interval id (count, min/avg/max/p95 duration); notes with `tid:{task_id}` pair per task
Task Lifecycle	Per-task `create`/`delete`/`suspend`/`resume` event summary from the `task` STI channel: created/deleted timestamps, suspend/resume counts, alive span (create→delete), total event count, and **Runs** (scheduler dispatch / context-switch-in count — distinct from Susp/Res, which only counts explicit `vTaskSuspend()`/`vTaskResume()` calls)
Core-Pair Migration Summary	Per directed core-pair migration count, lock-bounce count and percentage, and average off-CPU gap after migration; visible only on multi-core traces with migrations

Section	What it shows
Core Time Breakdown	Per-core split of span into **Active** (user tasks), **Idle** (IDLE task), **Tick** (TICK handler), and **Gap** (unaccounted time between segments)
Core Affinity	Per-task affinity mask history (`affinity_set` STI from `vTaskCoreAffinitySet`) vs. observed execution cores; **Violations** flags cores used while a restrictive mask was active (time-aware; shown only when those STI events are present)
Queue	Per-queue send/recv pairing by object pointer: hold count, issue count, average hold, and status (shown only when queue STI events are present)
Tag Analysis	Per `tag0_event`…`tag7_event` channel: sample count, min/avg/max/p95 of the tag value; click a row to open a scatter + histogram plot (shown only when tag STI samples are present)
Deadlines / CPU budget	Per-task slice violations (execution exceeding a configured nanosecond deadline) and CPU budget violations (task CPU% exceeding a global threshold); configure via **Settings → Display → Analysis thresholds** (`Ctrl+,`)

Core Migrations lists tasks that ran on two or more cores, with **Rate** (migrations per second of active time and per tick) and **Dwell** (average on-CPU slice length). For multi-core traces, open the **Migration heatmap** from the toolbar **Heatmap** button — click core-pair cells to drill into per-task sub-bins, then into Task View (see Migration heatmap); or open the **Migration chord diagram** from the toolbar **Chord** button for a directional overview of core-to-core migration volume (see Migration chord diagram). Toolbar **Analysis** flags thrashing / hot-pair candidates from the same stats. **Trace Compare…** (footer, next to Export) opens a dialog with **Summary**, **Top Tasks**, and **Core Migrations** tabs to diff two open trace tabs; optional cursor-range scoping compares each tab's C1–Cn window independently.

See Statistics metric tables for column definitions, distribution-chart usage, CDF overlay, and example plots from `tracedata/example-4cores.btf.gz`.

Snapshot Editor

Click the **Shot** toolbar button (or press S when focus is not in a text field) to capture the current timeline view and open the **Snapshot Editor**:

- Annotate with arrows, lines, double arrows, rectangles, circles, and text before exporting.
- Check **Dash** to draw dashed strokes (lines, arrows, rectangles, circles).
- Click a shape to select and drag it; **Ctrl+click** (or **Cmd+click** on macOS) duplicates the shape so you can place the copy.
- **Double-click** any shape to add or edit its label inline — text objects edit in place; lines and arrows place centered text aligned to the line; rectangles and circles place centered text.
- Single-click with the **Text** tool places a standalone text label inline.
- **Save PNG** or **Copy to Clipboard** from the editor footer.
- When the **Load** graph is visible, the capture includes the CPU load panel below the timeline.

Metrics Distribution Charts

In the **Statistics** panel, click any row in **Execution Time**, **Blocking Time**, **Inter-Arrival**, **Core Migrations**, **Preemption Chain**, **Priority Inheritance**, **Interval Analysis**, or **Tag Analysis** to open a floating chart popup. In **Trace Health (TICK)**, use the **Tick Distribution…** button (bar-chart icon beside the mode badge when ≥ 2 ticks are in scope). **Core Migrations** popups additionally show in-dialog tabs (**Dwell** / **Rate** / **Gap**) to switch metrics without closing the chart.

- **Scatter plot** — each event plotted in trace time order so you can spot trends, bursts, or outliers.
- **Histogram** — adaptive bar chart of the value distribution:
  - **Auto scale** (default) picks **linear**, **p5–p95**, or **log duration** from the data spread so bars are not squeezed when min/max or outliers span a wide range.
  - **Histogram scale** dropdown (Desktop toolbar above histogram; Web above the chart): **Auto**, **Linear**, **p5–p95**, **Log duration**.
  - **Adaptive bin count** (Freedman–Diaconis, 12–80 bins) instead of a fixed 50-bin linear split.

  - **Overflow buckets** in p5–p95 mode — separate dimmed bars for values below p5 and above p95, with counts in the caption.
  - **Log-scaled counts** when one bin dominates (tall spike vs many small bars).
  - **CDF overlay** — cumulative distribution curve on the histogram (see CDF overlay below).
  - Dashed reference lines for **avg**, **p50**, and **p95**; caption shows the active scale and full min–max range.
- **Export PNG / SVG** — buttons in the chart footer save the current scatter + histogram.

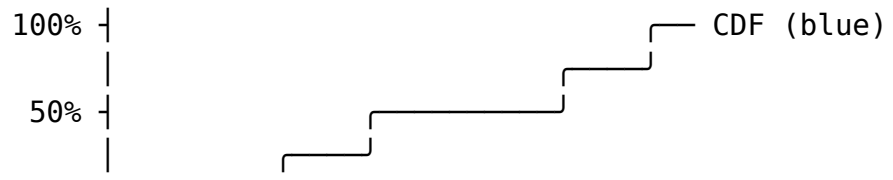
The popup can be dragged, resized, and closed independently of the main window. If the chart is open, it **updates live** when you move cursors or toggle cursor-range scope. Each browser tab keeps its own chart state when you switch between open traces.

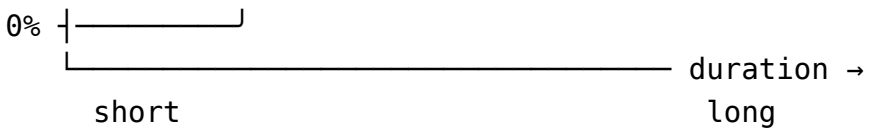
CDF overlay Every metrics histogram includes a **cumulative distribution function (CDF)** drawn as a **blue line** over the bars. It answers a different question than the histogram alone:

View	Question it answers
Histogram bars	*How many* samples fall in each duration bucket?
CDF curve	*What fraction* of samples are **at or below** a given duration?

The CDF is an **empirical** CDF (ECDF): for each sample in the active scope (full trace or cursor range), sorted by duration from shortest to longest, the curve plots **(duration → cumulative %)**. There is one step per sample; when several samples share the same duration, the curve rises vertically at that x position.

How to read the chart





- **Horizontal axis (bottom)** — duration (same scale as the histogram bars: linear, p5–p95, or log, depending on **Histogram scale**).
- **Left vertical axis** — sample **count** per bin (bar height).
- **Right vertical axis** — cumulative **percent** (0%, 50%, 100% labels on a dashed guide line).
- **Curve direction** — starts at the **bottom-left** (0% of samples below the shortest value) and rises toward the **top-right** (100% of samples included). A **steep** rise means many samples cluster at similar short durations; a **gradual** rise means values are spread out.

Relating the CDF to table columns and reference lines

The dashed vertical markers on the histogram are single-number summaries; the CDF shows the **full** cumulative picture:

Marker / table column	On the CDF
p50 (green)	Curve crosses **50%** on the right axis at the median duration.
p95 (orange)	Curve crosses **95%** — 95% of samples are shorter than this duration.
avg (purple)	Shown as a vertical line; the CDF does **not** pass through a fixed “avg %” because the mean is not a percentile.

Example readings (Execution Time for a task with 100 slices):

- At duration *D* where the CDF is at **30%**, roughly **30 slices** (30%) finished in *D* or less — useful for “how often does this task complete within my deadline?”

- If the curve is already above **90%** while still on the **left half** of the chart, most runs are short and the tail is light.
- If the curve stays below **50%** until far right, the distribution is **wide or skewed** — the histogram bars alone may look crowded; switch **Histogram scale** to **Log duration** or **p5–p95** and use the CDF to see where the bulk of the mass sits.

Histogram scale and the CDF

The CDF always reflects the **same samples** as the bars; only the **x mapping** changes with the scale mode:

Histogram scale	Effect on CDF
Auto / **Linear**	Duration mapped linearly from min to max.
p5–p95	Main curve spans the p5–p95 window; outliers appear in dimmed underflow / overflow buckets at the edges, and the CDF steps into those buckets at the corresponding percentiles.
Log duration	Short and long durations are spread across the axis; the CDF is easier to read when bars would otherwise pile up on the left.

The caption above the histogram (e.g. `log-scaled duration axis · full range 17 µs–975 µs`) always shows the **true** min–max range even when the axis is compressed or clipped.

When to use the CDF

- **Deadline / budget checks** — estimate what % of activations meet a time limit without reading individual scatter points.
- **Compare spread** — two tasks with similar **p50** can have very different CDF shapes (tight cluster vs long tail).

- **Skewed data** — after switching away from a crowded linear histogram, use the CDF with **p50** / **p95** lines to see how much of the population sits below each marker.
- **Cursor-scoped analysis** — with **Limit to cursor range** enabled, the CDF recalculates for only the slices inside C1–Cn, same as the table and scatter plot.

The CDF is included in **Export PNG / SVG** from the plot dialog. It is not interactive (no click-to-jump); use the **scatter plot** above the histogram to jump to individual events.

Jump links: in Execution Time, Blocking Time, and Inter-Arrival tables, click **Min** or **Max** (dotted underline) to jump to the slice at the shortest or longest value and add an **annotation** with a descriptive note. Click any **distribution-chart** point to jump to that event, add an annotation, and switch to the **Marks** tab with the new annotation selected (segment start for task metrics; tick timestamp for **Tick Distribution**; zoom + highlight for **Priority Inheritance** episodes; interval start for **Interval Analysis**). In Preemption Chain, the annotation is placed at the **preemptor segment** start. In **Mutex / Semaphore**, click any **Pairing issues** row to zoom to the running task segment on that core, jump to the issue time, and add an annotation.

Example plots from `tracedata/example-4cores.btf.gz` (4-core SMP trace, 67 tasks) are in Statistics metric tables.

STI events

Coloured diamond markers are shown on dedicated STI rows. Hover a marker for a tooltip showing the time, channel, event name, and note.

Interval markers (`interval_start` / `interval_stop`) are paired into measurable spans and drawn as horizontal bars on **Interval N** rows below the STI section (task view, horizontal orientation). When the BTF **note** includes `tid:{task_id}` (current FreeRTOS trace firmware), start/stop pair by **interval id + task id**; legacy traces without `tid` pair by the note string only. Raw start/stop marker channels are hidden from the STI row list. See Interval Analysis for pairing rules, statistics vs timeline behaviour, and limitations.

Status bar

Shows the number of tasks, segments, STI events, and total trace duration once a file is loaded.

Statistics & metrics

Shared by the Desktop **Statistics** tab and the Web **Statistics** panel.

The **Statistics** tab in the right-side panel (Desktop dock + Web tab) shows per-core CPU utilisation, top tasks, scheduling summary, trace health, and collapsible metric tables. Toggle visibility from **Settings → Display → Statistics panel**. For a quick triage of the same scope, use toolbar **Analysis** (see Analysis Findings) — findings are **not** listed as a panel section.

At the top, **Limit to cursor range (C1–Cn)** restricts all statistics (and Analysis Findings) to the time window from the first placed cursor through the last (requires 2+ cursors). Section titles show **(cursor range)** when scoped. Clearing all cursors returns to full-trace statistics immediately.

Layout: metric tables are **collapsible** (click a section title) and **resizable** (drag the thin handle below each table). On Desktop, drag the splitter between the timeline and CPU load graph to resize that pane; sizes are saved in `btf_viewer.rc`. On Web, right-panel width and stats table heights persist in `localStorage`.

It shows (in panel order):

- **Summary** — span, task/segment/STI counts (scoped when the checkbox is on)
- **Scheduling summary** — context-switch count and average/max core gap between consecutive slices on each core
- **Core utilisation** — percentage of active (non-IDLE, non-TICK) CPU time per core; **Load Balance Score** gauge at the top (score, σ, Gini; red / amber / green zones) (collapsible)
- **Core Time Breakdown** — per-core time budget split into **Active** (non-IDLE, non-TICK tasks), **Idle** (IDLE task), **Tick** (TICK handler), and **Gap** (unaccounted span between segments — scheduler latency / ISR overhead) expressed as percentages (collapsible)

- **Top tasks by CPU** — ranked list of worker tasks by total CPU time consumed (collapsible)
- **Trace health (TICK)** — tick period regularity, **tick / tickless mode detection** (coefficient of variation of tick intervals), large gaps, missed-tick estimate, and **Tick Distribution…** chart button (bar-chart icon beside the mode badge when ≥ 2 ticks in scope) (collapsible)
- **Core Migrations** — per-task migration count, **migration rate** (normalized per second of active time and per scheduler tick), **average core dwell time**, core count, primary core (% time), ping-pong count, STI events near migrations, and average off-CPU gap after migration vs other gaps; click a row for a distribution chart with **Dwell** / **Rate** / **Gap** tabs (collapsible)
- **Core-Pair Migration Summary** — per directed core-pair (e.g. `Core_0` → `Core_1`): total migration count, lock-bounce count and percentage, and average off-CPU gap after migration (collapsible; shown only on multi-core traces with migrations)
- **Execution Time Per Slice** — per-task min/avg/max/p95, run count, and CPU%; click a row for a scatter + histogram popup; click **Min** / **Max** to jump and annotate the BCET / WCET slice
- **Blocking Time** — off-CPU gap between consecutive activations of the same task (**Response Time** in Tracealyzer; same value, different label); min/avg/max/p95; click a row for a distribution chart; click **Min** / **Max** to jump and annotate the shortest / longest off-CPU gap (collapsible)
- **Inter-Arrival Time** — same statistics for gaps between task activations; click **Min** / **Max** to jump and annotate the shortest / longest inter-arrival gap (collapsible)
- **Preemption Chain Analysis** — for each victim/preemptor pair: count, total/average/max preemption overlap; click a row for a distribution chart; click a scatter point to jump and add an annotation at the preemptor segment (collapsible)
- **Priority Inheritance** — per-task base/peak priority, boost episodes, boosted time, and pattern (mutex inherit / L/M/H / boost only); click a row for a duration plot; click a scatter point to zoom, highlight, and annotate the episode (collapsible; shown only when the trace contains priority-inheritance STI events)
- **Mutex / Semaphore pairing** — per-object hold count, issue count, **core bounce count**, average hold, and status; **Pairing issues** sub-table lists orphan gives, cross-task gives, unmatched takes, teardown warnings, and **`CORE_MIGRATION_WHILE_HELD`** warnings (lock that crossed core boundaries while held); click an issue row to zoom, jump, and annotate (collapsible; shown

only when the trace contains sync-object STI events)

- **Queue** — per-queue send/recv pairing by object pointer: hold count, issue count, average hold, and status (collapsible; shown only when the trace contains queue STI events)
- **Task Lifecycle** — per-task summary of `create`, `delete`, `suspend`, and `resume` events recorded on the `task` STI channel: created/deleted timestamps, suspend/resume counts, alive span (create→delete), total lifecycle event count, and **Runs** — the number of times the task was actually dispatched onto a core (context-switch-in / segment count). Runs is a scheduler-level metric and is normally much larger than Susp/Res, which only counts explicit `vTaskSuspend()`/`vTaskResume()` API calls — a task can run (and be preempted/resumed by the scheduler) many times without ever being suspended. In `example-8cores.btf.gz`, test 9 creates **SR0–SR3** (pinned across cores) with overlapping suspend-while-blocked and suspend-while-running rounds — use this section to confirm Susp/Res counts match the STI pairs (collapsible; shown only when the trace contains task lifecycle STI events)
- **Core Affinity** — per-task affinity mask history from `affinity_set` STI (`vTaskCoreAffinitySet`) vs. observed execution cores; **Violations** lists cores used while outside the mask then in effect (slices before the first set are unrestricted; mask changes are shown as `0x1 → 0x8`) (collapsible; shown only when those STI events are present)
- **Deadlines / CPU budget** — configurable per-task execution deadline (nanoseconds) and global CPU budget threshold (%); shows **Slice over deadline** violations (task execution slices longer than the per-task limit, sorted by excess) and **CPU budget exceeded** violations (tasks whose CPU% in the current scope exceeds the budget); configure via **Settings → Display → Analysis thresholds** (`Ctrl+,`) — the section is always visible and displays a prompt to open Settings when no thresholds are configured (collapsible)
- **Interval Analysis** — per interval id: count, min/avg/max/p95 duration of paired start→stop spans; pairing uses `tid` in the note when present; click a row for a duration plot; click a scatter point to jump and add an annotation at the interval start (collapsible)
- **Tag Analysis** — per `tag0_event`…`tag7_event` STI channel: sample count, min/avg/max/p95 of the tag value; click a row to open a scatter + histogram plot (collapsible; shown only when the trace contains tag STI samples)

Export CSV / **Export HTML** respect the current cursor scope. **Export CSV** includes summary tables for every statistics section and a **Core Affinity Violations**

sub-table listing every mutex that crossed core boundaries, plus **Load Balance Score**, σ, and Gini coefficient under Core Utilisation. **Export HTML** adds the same summaries plus an **Analysis Findings** card near the top (same heuristics as toolbar **Analysis** — load imbalance, WCET/CPU hotspots, blocking, L/M/H priority inversion, core thrashing / hot pairs, deadline breaches, tick health, sync/mutex bounces), embeds the **Load Balance Score** gauge as a self-contained SVG <img> (data URI) under Core Utilisation, and detail sub-tables for **Priority Inheritance** (boost episodes), **Mutex / Semaphore** (pairing issues with bounce warnings, and hold episodes with take/give core columns), and **Interval Analysis** (individual instances). **Trace Compare…** compares summary, top tasks, and core migrations between two open tabs; enable **Limit to each tab's cursor range** to scope each side to its own C1–Cn window. Open metrics charts update live when cursors move or scope is toggled; each trace tab remembers its own open chart when you switch tabs.

Full column definitions, chart axis meanings, and example plots: Statistics metric tables.

Statistics metric tables

The Statistics panel (Desktop **Statistics** tab + Web **Statistics** tab) organises metrics into collapsible sections. Tables are **sortable** — click a column header to sort ascending/descending. **Export CSV** and **Export HTML** at the panel footer honour the current cursor scope and include every section's summary table. **Export HTML** starts with an **Analysis Findings** card (same content as toolbar **Analysis** — load balance, WCET, blocking, thrashing, deadlines, tick health, and sync) and embeds the **Load Balance Score** gauge as an SVG image under Core Utilisation; use the toolbar dialog for interactive triage and **Save as text…**. HTML also adds detail sub-tables under Priority Inheritance, Mutex / Semaphore, and Interval Analysis (longest instances / hold episodes first, capped at 150–200 rows per sub-table).

How to use the panel

1. Open a trace (e.g. `tracedata/example-4cores.btf.gz` for a 4-core SMP workload, or `tracedata/example-2cores.btf.gz` for a smaller 2-core demo).
2. Optionally click toolbar **Analysis** for a severity-tagged triage of the current scope.
3. Expand the sections you care about (or use the **+** / **−** icons at the top to expand/collapse all).

4. Optionally place **2+ cursors** and enable **Limit to cursor range (C1–Cn)** to restrict every metric (and Analysis Findings) to a time window.
5. Click a **table row** to open a distribution chart (where supported), click **Min** / **Max** to jump and add an annotation at an extreme slice on the timeline, or click a **Mutex / Semaphore** issue row to zoom, jump, and annotate at that STI event.
6. Use **Trace Compare…** when two traces are open to diff summary and migration stats.

The example plots below were generated from **`tracedata/example-8cores.btf.gz`** (8 cores, 154 tasks, ~31 100 segments, time scale us). Regenerate them with:

```
make -C BTFViewer update-images
```

This invokes the desktop CLI `snapshot` command (`--view plot` / `--view timeline`) directly — see `make -C BTFViewer help`. Timeline screenshots (e.g. `images/stats/tasks-priority-low.svg`) use `--view timeline --task ... --lo ... --hi ...` to zoom to the region of interest (the range is applied **after** the offscreen view is laid out so the exported image fills that window). Multi-core migration illustrations (e.g. `images/stats/tasks-migrate-cs22.svg`) add `--view-mode core` so the highlighted task is shown hopping across expanded core rows. Fit-to-window + per-core task CPU Load illustrations (e.g. `images/stats/tasks-cpu-load-cs22.svg`) use `--view-mode core --task ... --cpu-load` with no `--lo`/`--hi`. Migration heatmap screenshots (`images/heatmap-pairs.svg`, `images/heatmap-tasks.svg`) are regenerated the same way with `make -C BTFViewer update-images` (`snapshot --view heatmap`, with `--drill-row`/`--drill-bin` for the task-level image) — see Migration heatmap. The migration chord diagram screenshot (`images/migration-chord.svg`) is regenerated with the same target (`snapshot --view chord`) — see Migration chord diagram.

Summary, scheduling, and core utilisation These sections appear at the top of the Statistics panel (not sortable tables).

Summary — scope-wide counts: trace span, tasks, segments, STI events. Span is *t*max − *t*min in the active scope (full trace or cursor range).

Scheduling summary — per core, **context switches** count slice boundaries and **core gap** is idle time between consecutive slices on that core:

$$g_{\text{core}} = t_{\text{start},k+1} - t_{\text{end},k}$$

Large **max core gap** on a core that should be busy suggests starvation, tickless idle, or a single long-running task blocking others.

Core utilisation — per core, share of non-IDLE, non-TICK active time in scope:

$$U_{\text{core}} = \frac{T_{\text{active,core}}}{T_{\text{scope}}} \times 100$$

When two or more cores are present, a **Load Balance Score** gauge is shown at the top of the section:

$$\text{Score} = 100\% \times (1 - G), \quad G = \text{Gini coefficient of } \{U_{\text{core}}\}$$

The gauge needle shows the score (100 = perfect balance, 0 = single-core overload). Zones match toolbar **Analysis**:

Zone	Condition	UI
Red	Score < 70 %	**Unbalanced** chip + alert (red zone)
Amber	Score ≥ 70 % and σ > 30 %	**σ > 30%** chip
OK (green)	Score ≥ 70 % and σ ≤ 30 %	Green score

Toolbar **Analysis** also warns when Score < 70 % *or* σ > 30 %, and only describes cores as “reasonably balanced” when Score ≥ 85 % and σ ≤ 30 %. **Export HTML** embeds the gauge as a self-contained SVG <img> (data URI) under Core Utilisation; **Export CSV** includes the score, σ, and G values.

What it tells you: Imbalanced utilisation across cores may indicate poor affinity, lock pinning, or workload placement issues — cross-check with **Core Migrations**, the migration heatmap, and toolbar **Analysis**.

Top tasks by CPU — ranks tasks by *T*exec,i (same denominator as **CPU%** in Execution Time Per Slice).

Execution Time Per Slice Measures how long each **on-CPU slice** lasts for a task — from switch-in until the task blocks, yields, or is preempted.

Formula — for task *i*, each on-CPU slice *k* in scope:

$$d_k = t_{\text{end},k} - t_{\text{start},k}$$

Table statistics (Min / Avg / Max / p95) are computed over all slice durations *d*k in scope. **CPU%** is the task's share of total active CPU time in scope:

$$\text{CPU}_i = \frac{T_{\text{exec},i}}{\sum_j T_{\text{exec},j}} \times 100$$

What it tells you: Short, uniform slices suggest periodic or tick-driven scheduling. A long **Max** or heavy **p95** tail marks worst-case execution time (WCET) slices — often critical sections, lock holds, or interrupt-disabled regions. Compare **Min** (BCET) and **Max** (WCET) to judge jitter; a wide spread on a real-time task may violate deadline assumptions even when **Avg** looks acceptable.

Column	Meaning
Task	Display name (`Name[id]`)
Runs	Number of slices in scope
CPU%	Share of total trace (or cursor-range) active time
Min / **Avg** / **Max** / **p95**	Slice duration statistics
Min / **Max** links	Jump and annotate BCET / WCET slice

Distribution chart — click any row:

- **Scatter:** x = slice start time, y = slice duration.
- **Histogram:** distribution of slice durations (auto log scale when the tail is wide; see CDF overlay for reading the blue cumulative curve).

In `example-8cores.btf.gz`, task **CS[11]** has 730 slices with a long tail of longer runs (context-switch stress tasks):

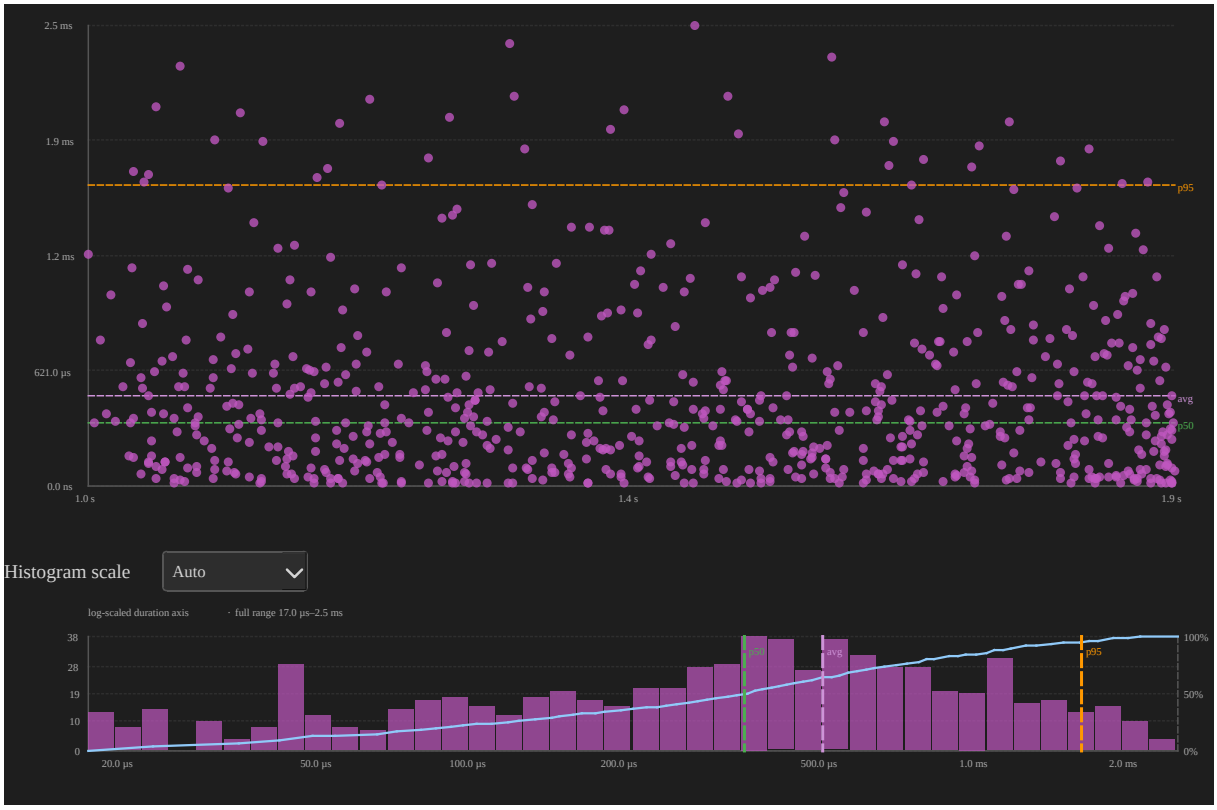


Figure 4: Execution time distribution for CS[11] in example-8cores.btf.gz

The scatter shows periodic bursts of short slices; the histogram uses a **log-scaled duration axis** so short and long slices are both visible. The **CDF** rises steeply on the left (most slices are short) then levels toward 100% as longer runs are included; **p50** and **p95** vertical markers align with the 50% and 95% ticks on the right axis.

Blocking Time Measures the **off-CPU gap** between the end of one slice and the start of the next for the same task — time spent waiting to run again (preempted, blocked on a resource, or delayed by the scheduler).

> **Name alias — Response Time:** Percepio Tracealyzer and similar tools call this metric **Response Time** (time from the end of one task activation to the start of the next). BTFViewer labels it **Blocking Time** only; the statistic, formula, and charts are the same — there is no separate Response Time row in the UI.

Formula — for consecutive activations k and $k+1$ of task i:

$$g_k = t_{\text{start},k+1} - t_{\text{end},k}$$

Only positive gaps are counted. Min / Avg / Max / p95 are taken over all gaps gk in scope.

What it tells you: Blocking time is pure **wait time** — the task is runnable or blocked but not on-CPU. High **Avg** or **Max** gaps often point to lock contention, priority inversion, or a higher-priority task monopolizing the core. Spikes clustered at specific times in the scatter plot usually correlate with a particular preemptor or synchronization object; use **Preemption Chain Analysis** or **Mutex / Semaphore** pairing to find the cause.

Column	Meaning
Task	Display name
Gaps	Number of positive off-CPU gaps
Min / Avg / Max / p95	Gap duration statistics
Min / Max links	Jump and annotate resume slice at shortest / longest gap

Distribution chart — click any row:

- **Scatter:** x = resume time, y = off-CPU gap.
- **Histogram:** distribution of blocking gaps.

CS[11] in `example-8cores.btf.gz` (729 gaps):

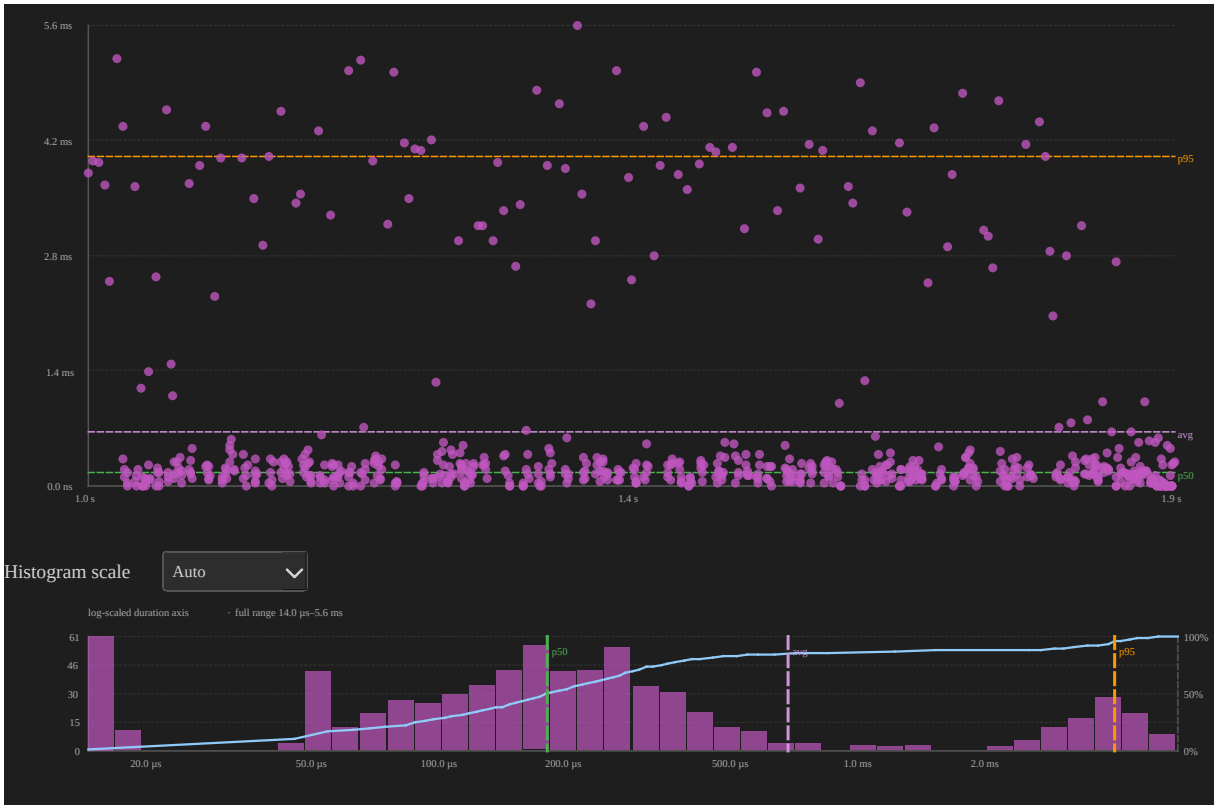


Figure 5: Blocking time distribution for CS[11] in example-8cores.btf.gz

High blocking gaps clustered at certain times often correlate with lock contention or a higher-priority task dominating the core.

Inter-Arrival Time Measures the gap between **successive activation start times** of the same task (time between slice starts, not off-CPU gap).

Formula — for consecutive activations *k* and *k+1*:

$$\Delta t_k = t_{\text{start},k+1} - t_{\text{start},k}$$

Min / Avg / Max / p95 are taken over all inter-arrival samples Δ*t*k in scope.

What it tells you: Inter-arrival time reflects how often the task is **scheduled to run**, including time it spent on-CPU. For periodic tasks it should cluster near the expected period; drift or bimodality hints at missed deadlines, timer jitter, or workload-dependent release patterns. Because Δ*t*k = *d*k + *g*k (slice duration plus blocking

gap), inter-arrival is always ≥ blocking time for the same activations — compare both tables to see whether jitter comes from short runs or long waits.

Column	Meaning
Task	Display name
Runs	Number of inter-arrival samples
Min / **Avg** / **Max** / **p95**	Gap between activation starts
Min / **Max** links	Jump and annotate activation at shortest / longest inter-arrival

Distribution chart — click any row:

- **Scatter:** x = activation time, y = gap since previous activation.
- **Histogram:** distribution of inter-arrival gaps.

CS[11] in `example-8cores.btf.gz`:

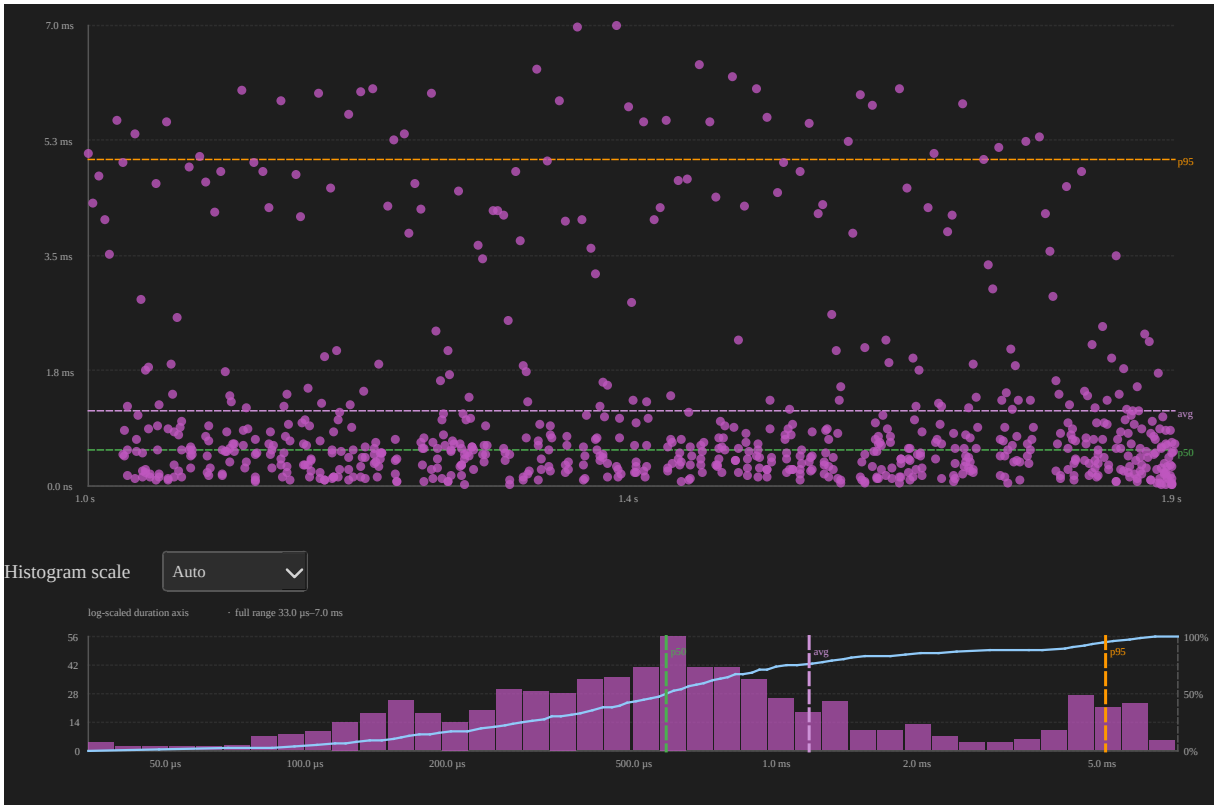


Figure 6: Inter-arrival time distribution for CS[11] in example-8cores.btf.gz

Compare with Blocking Time (Response Time): inter-arrival includes time the task was **running**, so values are typically larger than off-CPU gaps alone. The histogram auto-selects **log duration** for CS[11] because activation gaps span microseconds to milliseconds.

Preemption Chain Analysis For each **victim** task's off-CPU gap, the analyser finds which **preemptor** tasks ran on the **same core** as the victim during that gap and aggregates overlap duration.

Formula — for victim v and preemptor p on the same core during gap g:

$$\text{overlap}(v, p, g) = \sum_{p \in g} [\min(t_{\text{end}}, g_{\text{end}}) - \max(t_{\text{start}}, g_{\text{start}})]$$

Count is the number of such overlap events; **Total** / **Avg** / **Max** summarise overlap durations for each victim←preemptor pair.

What it tells you: Answers *who ran while this task waited*. A single pair with high **Count** and **Total** means one preemptor dominates the victim's blocking; many pairs with moderate counts suggest fair sharing or frequent context switches. Large **Max** overlap points to long stretches where the victim was ready but could not run — a common sign of priority misconfiguration or a CPU hog.

Column	Meaning
Victim	Task that was off-CPU
Preemptor	Task that ran during the victim's gap
Count	Number of preemption overlap events
Total / Avg / Max	Overlap duration (how long the preemptor held the CPU during victim gaps)

Distribution chart — click any row (victim ← preemptor pair):

- **Scatter:** x = when the preemption overlap started, y = overlap duration.
- **Histogram:** how long that preemptor typically held the CPU during gaps.
- **Click a point** to jump to the **preemptor's segment** and add an annotation with duration/time notes.

CS[24] ← CS[25] in `example-8cores.btf.gz` (55 overlap events, 14.936 ms total overlap) — two context-switch stress tasks repeatedly preempting each other:

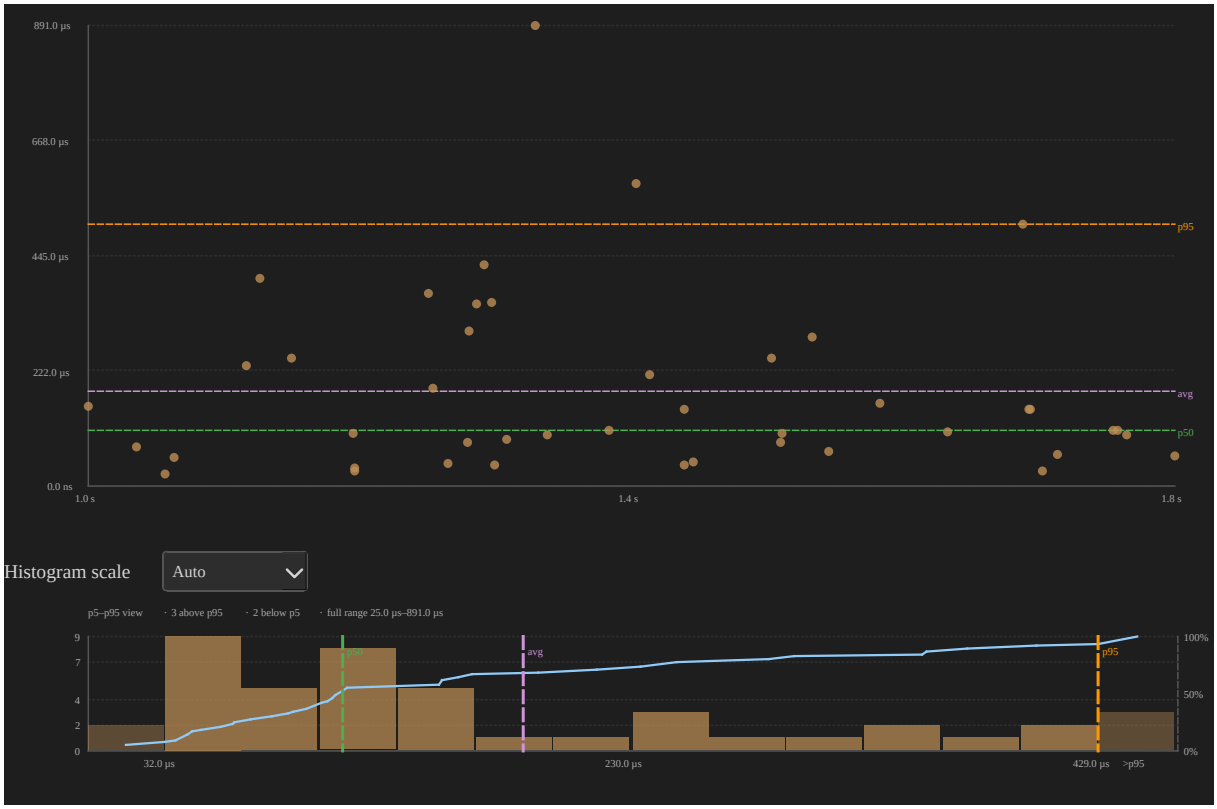


Figure 7: Preemption chain distribution CS[24] preempted by CS[25] in example-8cores.btf.gz

High **Count** with moderate **Avg** overlap suggests frequent short preemptions; a few points with large **y** values are long stretches where CS[25] ran while CS[24] waited. Use this table to answer *who preempted whom* and whether a victim's blocking is dominated by one preemptor or many.

Priority Inheritance Shown when the trace has **`create pri:N`** on task-create T rows **and** at least one priority STI on the `task` channel:

STI note prefix	Hook	Meaning
`priority_inherit Name[id] pri:N`	`traceTASK_PRIORITY_INHERIT`	Mutex holder inherited priority *N*

STI note prefix	Hook	Meaning
`priority_disinherit Name[id] pri:N`	`traceTASK_PRIORITY_DISINHERIT`	Mutex holder returned to base priority *N*
`set_priority Name[id] pri:N`	`traceTASK_PRIORITY_SET`	Explicit `vTaskPrioritySet()`

Formula — a **boost episode** is a contiguous interval where the task's effective priority is above its **Base** (from `create pri:N`):

$$T_{\text{boosted}} = \sum_{\text{episodes}} (t_{\text{end}} - t_{\text{start}})$$

where *t*start is the inherit / set-up STI and *t*end is the disinherit / set-back STI for each episode.

Boosts counts episodes; **Peak** is the highest priority observed while boosted.

What it tells you: Priority inheritance prevents priority inversion when a low-priority mutex holder blocks a high-priority waiter. Long **Boosted** time or many **Boosts** on a task that is not the mutex owner under test may indicate lock contention or chained inheritance. **Mutex inherit** confirms the kernel raised priority via `priority_inherit`; **L/M/H pattern** flags the classic three-level inversion geometry (see below); **Boost only** means manual `vTaskPrioritySet()` without mutex hooks.

Column	Meaning
Task	Mutex holder or subject whose priority was raised
Base	Priority at task create (`create pri:N`)
Peak	Highest `set_priority` level observed while boosted
Boosts	Number of boost episodes (base → above base → back to base)

Column	Meaning
Boosted	Total time above base priority in scope
Pattern	Summary label — see L/M/H pattern and table below

Pattern (summary)	When it appears
Mutex inherit	At least one boost episode used `priority_inherit` / `priority_disinherit`, and no extra inversion episodes beyond inherits
Mutex inherit + L/M/H	Mutex inheritance **and** additional boost episodes where medium-priority tasks sit between base and peak
L/M/H pattern	Boost above base (usually `set_priority`) with medium-priority tasks between **Base** and **Peak**, but no `priority_inherit` on that task
Boost only	Raised above base with no medium tasks in range and no mutex inherit

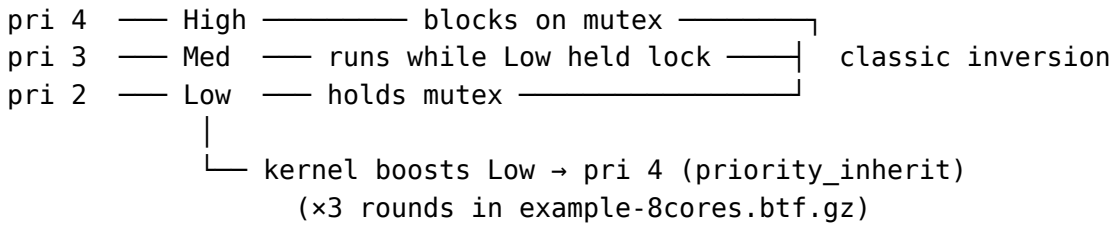
Per-episode detail (distribution chart tooltips, **Export HTML → Boost episodes**) can be more specific, e.g. `Mutex inherit L/M/H (CS[11], CS[12] +126)` — listing up to two medium-task names plus a count of any others.

L/M/H pattern (priority inversion geometry) **L/M/H** names the textbook **priority-inversion** layout on three priority levels. In the demo firmware (`Demo/examples/freertos_test/main.c` test 8) the tasks are named **Low**, **Med**, and **High**, pinned to **Core_0** on SMP, and the scenario is repeated **T8_ROUNDS (3)**

times so the timeline shows multiple red boost stripes and the Priority Inheritance chart has several inherit points:

Role	Meaning	In `example-8cores.btf.gz` test 8
L (Low)	Mutex holder at the lowest priority of the three	**Low[266]** — `create pri:2`, holds the test mutex
M (Medium)	Runnable work at a priority **between** L and H; can preempt L while H waits	**Med[267]** — `create pri:3`, CPU work after Low takes the mutex
H (High)	Blocked on the mutex L holds; triggers inheritance	**High[268]** — `create pri:4`, blocks on the same mutex

Without mutex priority inheritance, **M** would run while **H** waited on **L** — unbounded inversion. FreeRTOS boosts **L** to **H**'s priority (e.g. first `priority_inherit Low[266] pri:4` near ~3099 ms) so **L** finishes its critical section before **M** can starve **H**.



How the viewer detects L/M/H: at the end of each boost episode, it scans every task with a known `create pri:N`. Any task whose **base priority** satisfies **Base** < *p* < **Peak** (strictly between the boosted task's base and peak) is a **medium blocker**. If at least one exists, the episode is an **inversion suspect** and contributes to the **L/M/H** pattern.

- **Episode** level: `inversionSuspect` when the episode was mutex-inherited **or** medium blockers exist; episode **Pattern** may list medium task names.

- **Summary** level (table **Pattern** column): aggregates episodes — `Mutex inherit`, `L/M/H pattern`, `Mutex inherit + L/M/H`, or `Boost only`.

Important: medium-blocker detection uses **all** tasks in the trace with recorded create priorities, not only tasks active in that instant. On a busy SMP trace (many workers at `pri:3` from earlier tests), the episode label may show several medium names (e.g. `Mutex inherit L/M/H (CS[11], CS[12] +126)`). For test 8, **Med[267]** is the semantically relevant **M**; cross-check with the timeline around test 8 (~3042–3359 ms absolute, `--lo 3042000 --hi 3359000`) and `Demo/examples/freertos_test/main.c` test 8 (`vInvLow` / `vInvMed` / `vInvHigh`).

Example — test 8 in `example-8cores.btf.gz` (`Low[266]`):

Field	Value
Base / Peak	2 → 4
Boosts / Boosted	**3** mutex-inherit episodes, ~34.4 ms each (~103 ms total)
Summary Pattern	**Mutex inherit**
Episode Pattern	**Mutex inherit L/M/H** (medium tasks include **Med[267]** at pri 3)
STI windows	`3099448→3133873`, `3187403→3221850`, `3275380→3309826` µs

Zoom the timeline to that window (or click a **Low[266]** scatter point) to see the **three red** boost stripes on the Low row and **High** blocking on the mutex between them.

Contrast — test 7 (`PS[228]`, manual boost): the runner calls `vTaskPrioritySet(subject, BOOST_PRIORITY)` — trace shows `set_priority` STI events, not `priority_inherit`. **PS[228]** has the same numeric geometry (base 2, peak 4, medium fillers at pri 3) so the summary **Pattern** is **L/M/H pattern** (not **Mutex inherit**). Boost duration is much shorter (~119 µs) because no mutex hold loop is involved. Use **PS[228]** vs **Low[266]** to separate kernel inheritance from application-driven priority changes.

Timeline UX — boosted periods appear as a **bottom stripe** on the task row (horizontal) or a **right-edge stripe** (vertical): **orange** = boost only (`Boost only` /

manual `set_priority` without L/M/H geometry); **red** = mutex inherit or any L/M/H-related pattern. Task labels show **`· pri N`** when create priority is known.

Low[266] on the timeline in `example-8cores.btf.gz` (zoomed to test 8, `--lo 3042000 --hi 3359000`). Three **red bottom stripes** on the Low row mark the `priority_inherit` → `priority_disinherit` windows (~34.4 ms each). **Med[267]** runs on the same core before each inherit; during each boost window Med stays off-CPU:

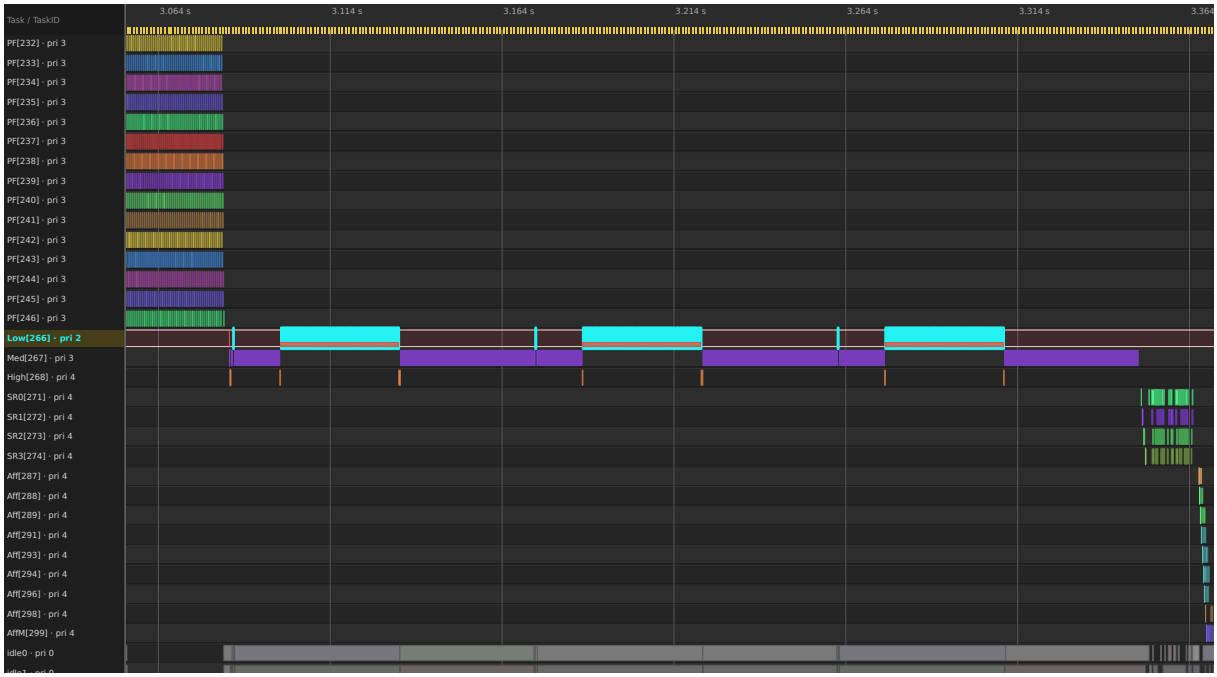


Figure 8: Timeline view: Low[266] with three red priority-inheritance stripes (example-8cores.btf.gz)

Export a timeline SVG from the viewer (**File → Save SVG** or toolbar) after zooming to the episode, or regenerate it headlessly via `make -C BTFViewer update-images` (desktop CLI `snapshot --view timeline --task ... --lo ... --hi ...`).

Distribution chart — click any Priority Inheritance table row:

- **Scatter:** x = boost episode end time, y = boosted duration. Orange points = boost only; red = L/M/H / mutex inherit. Test 8 contributes **three** clustered red points near ~34 ms.
- **Histogram:** distribution of boost durations.

- **Click a point** to zoom to that episode, scroll to the task row, highlight it, and add an annotation (re-click skips duplicate annotations).

Export HTML includes a **Boost episodes** detail sub-table (up to 200 rows by start time).

Low[266] distribution (test 8: three mutex inherit episodes, base pri 2 → peak 4, ~34.4 ms each):

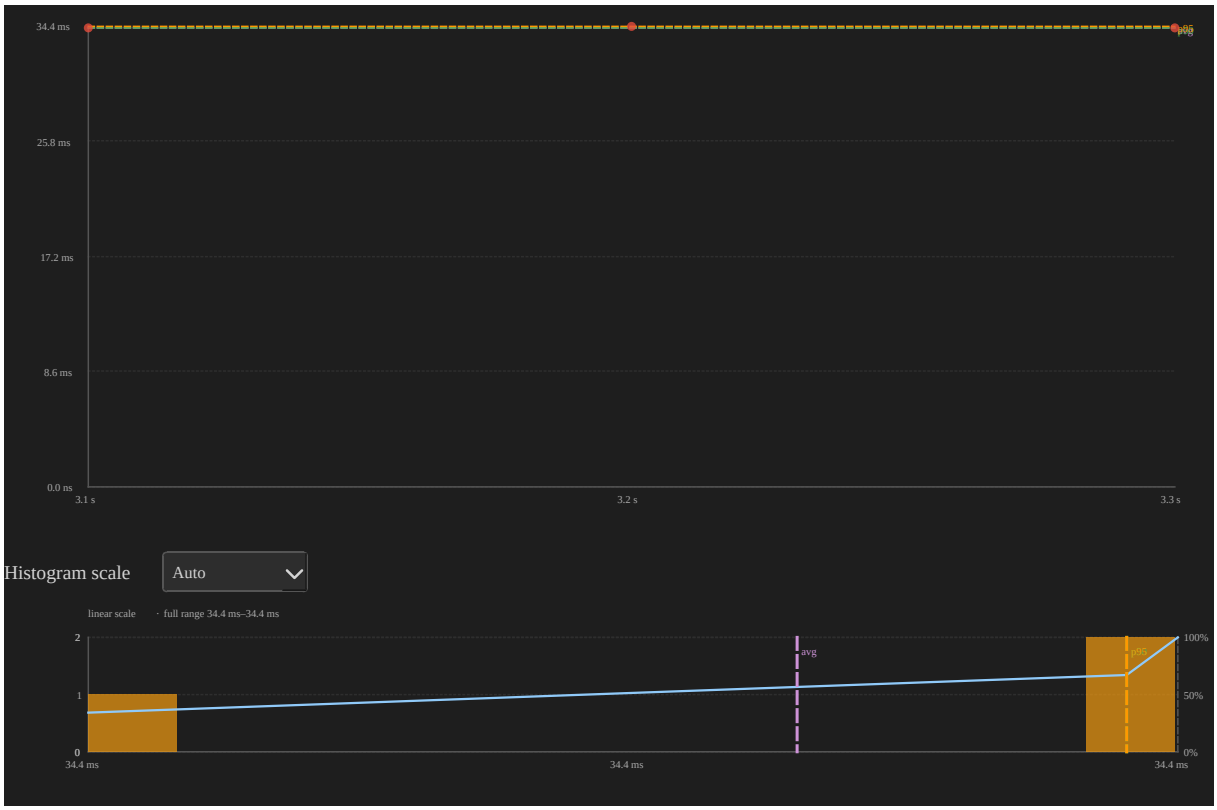


Figure 9: Priority boost distribution chart for Low[266] in example-8cores.btf.gz

PS[228] (test 7: manual `vTaskPrioritySet`, **L/M/H pattern**, ~119 µs boosted) contrasts with **Low[266]** above — same base/peak numbers, different mechanism and duration.

Note: `traceTASK_PRIORITY_INHERIT` / `traceTASK_PRIORITY_DISINHERIT` are invoked by the FreeRTOS kernel inside `xTaskPriorityInherit()` / `xTaskPriorityDisinherit()` when `configUSE_MUTEXES` is enabled.

Mutex / Semaphore pairing When queue trace hooks are enabled (`configINCLUDE_QUEUE_EV` mutex and semaphore STI lines include the **FreeRTOS object pointer** in the note:

```
703266,Core_0,0,STI,mutex,0,trigger,take 0x80018700
707222,Core_3,0,STI,mutex,0,trigger,give 0x80018700
```

The viewer pairs **`take`/`give`** (and **`create`/`delete`**) **per pointer** — not per STI channel alone — so two mutexes of the same type stay distinct. The kernel **`give`** immediately after **`create`** (mutex / binary sem available) is ignored when it falls within **1 ms** of the create event.

Semaphores use two pairing directions automatically:

Pattern	Example	Pairing
Hold (take → give)	Area slot sem, worker acquires then releases	`take` opens, matching `give` closes (FIFO)
Signal (give → take)	*_done / *_go coordination sems	`give` posts, matching `take` consumes (FIFO)

Mutexes always use hold pairing (LIFO — owner must `give`).

Formula — for each paired hold span h of duration τh:

$$\overline{\tau}_{\text{hold}} = \frac{1}{N_{\text{holds}}} \sum_{h} \tau_h$$

What it tells you: Avg hold shows typical lock or semaphore residency time — very long holds inflate blocking for waiters. **Issues** > 0 (orphan give, cross-task give, unmatched take, delete while held) mean the trace does not form clean take/give pairs and hold statistics may be incomplete. **Deadlock risk** at trace end flags multiple mutexes still held by different tasks — verify whether that is expected teardown or a real stall.

Column	Meaning
Object	Kind + pointer (`mutex 0x80018700`)

Column	Meaning
Kind	`mutex` or `sem`
Holds	Number of paired take→give spans in scope
Issues	Pairing problems in scope
Avg hold	Mean hold duration across paired spans
Status	**OK**, **Warning**, or **Error**

Check	Severity	Meaning
Orphan give	Error	Mutex `give` with no matching open `take` on that pointer
Cross-task give	Warning	Mutex `give` by a different task than the one that `take`
Unmatched take	Warning	`take` still open at trace end
Unmatched give	Warning	Semaphore `give` still unmatched at trace end
Delete while held	Warning	`delete` while resource `take`s are still open (common during teardown)

Check	Severity	Meaning
CORE_MIGRATION_WHILE_HELD	Warning	Mutex `take` on one core, `give` on a different core — the lock crossed a core boundary while held; indicates a **cache-line bounce** where the hardware cache line containing the lock data is transferred between cores. Shown in the Pairing Issues sub-table with the exact from/to cores (e.g. `Lock bounced from Core_0 to Core_1`)
Deadlock risk	Warning	≥2 mutexes still held by ≥2 different tasks at trace end

The running task for each event is inferred from the **core timeline** at that timestamp (same approach as interval `tid` pairing).

Below the summary table, a **Pairing issues** sub-table lists every problem in scope (time, object, issue kind, detail). **Click any issue row** to zoom to the running task segment on that core (when found), jump to the issue timestamp, highlight the segment, and add an annotation with a descriptive note (Desktop + Web). Re-clicking the same point skips duplicate annotations.

Export HTML adds two detail sub-tables under this section: all **Pairing issues** in scope (including `CORE_MIGRATION_WHILE_HELD` warnings), and **Hold episodes** (longest first, up to 150 rows) with **Take core** and **Give core** columns.

Export CSV includes a **Core Affinity Violations** sub-section listing each mutex that had at least one bounced hold, with the bounce count and a description.

`example-4cores.btf.gz` (tests 1–3) exercises `0x80018700` (mutex) and `0x80018650` (counting sem) with clean hold pairing. The full trace shows `CORE_MIGRATION_WHILE_HELD`

warnings on `0x80018700` (3 bounces) and `0x80018650` (1 bounce) — the Statistics **Mutex / Semaphore** summary table will show **Warning** for these mutexes and non-zero values in the **Bounces** column. Coordination sems pair in **signal** direction.

Interval Analysis Pairs **`interval_start` / `interval_stop`** STI events into measurable code regions. Each interval **id** gets an **Interval N** row on the timeline (horizontal task view) with colored span bars; the statistics table aggregates duration across all paired spans for that id.

Formula — for each paired instance *j* with start *t*s and stop *t*e:

$$\tau_j = t_e - t_s$$

Count is the number of paired spans in scope; Min / Avg / Max / p95 are over all interval durations τj.

What it tells you: Interval metrics measure **how long instrumented code regions take** — loop iterations, critical sections, or end-to-end handlers. Tight clusters in the distribution chart mean stable iteration time; outliers or a high **Max** often mark contention, preemption inside the region, or pairing artefacts (see **Limitations** under Interval Analysis below). Compare interval ids to separate workloads (e.g. mutex stress vs lighter loops in the same trace).

BTF note field (last CSV column on each interval line):

Format	Example	Viewer pairing
Current firmware	`1 tid:7` or `0 tid:0x8`	Interval id + task id (decimal or `0x` hex); same numeric id on different tasks does not cross-pair
Legacy	`1`	Full note string `1` only

Recorded by `traceINTERVAL_START(id)` / `traceINTERVAL_STOP(id)` in firmware — task id is captured automatically in `param2` and emitted in the note as `tid:…`. See Binary → BTF dump mapping in the repo root README.

Column	Meaning
ID	Interval id from `traceINTERVAL_START(id)` / `traceINTERVAL_STOP(id)`
Label	Display name (`Interval N`)
Count	Number of paired start→stop spans in scope
Min / **Avg** / **Max** / **p95**	Interval duration statistics

Distribution chart — click any row:

- **Scatter:** x = interval stop time, y = interval duration.
- **Histogram:** distribution of interval durations.
- **Click a point** to jump to the **interval start** and add an annotation with duration/time notes.

Export HTML includes an **Interval instances** detail sub-table (longest first, up to 200 rows).

Interval 1 in `example-8cores.btf.gz` (1728 spans from eighteen `vCtxSwitchWorker` tasks — CS[11]–CS[28] — sharing id 1 — each worker pairs via its own `tid` in the note):

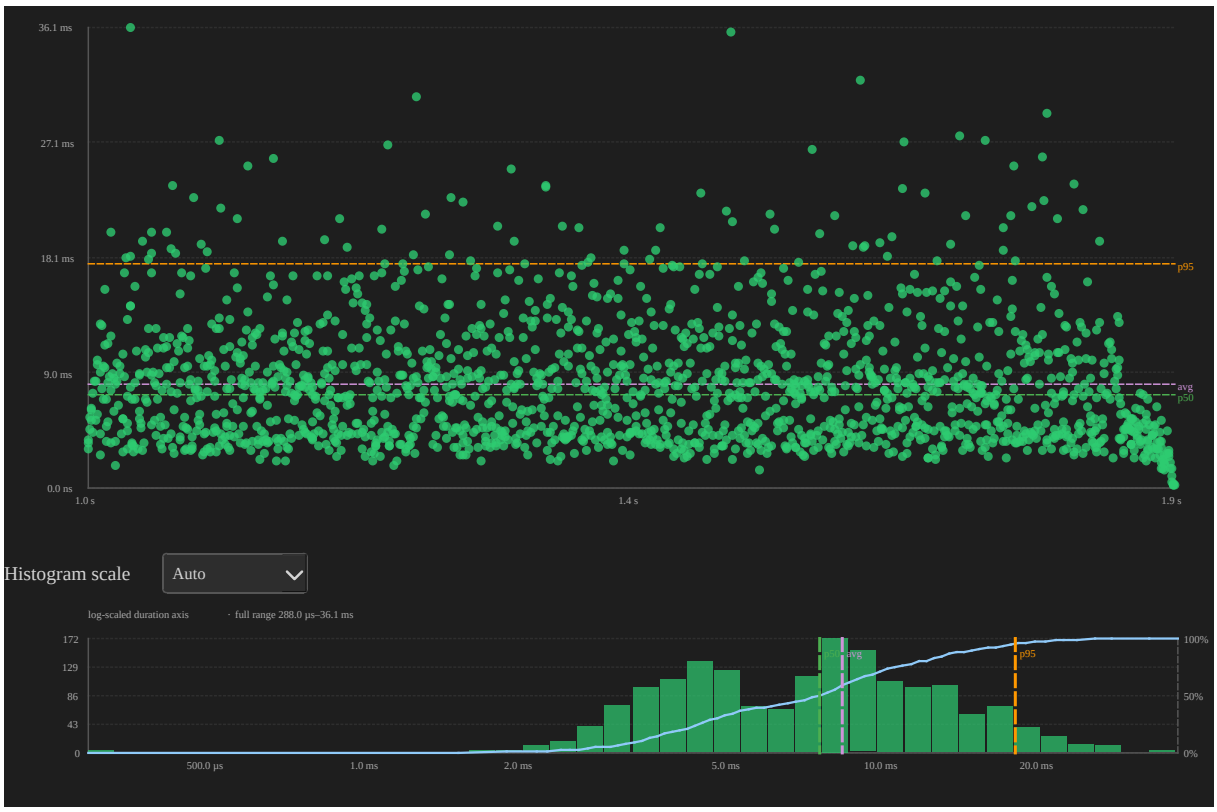


Figure 10: Interval duration distribution for interval id 1 in example-8cores.btf.gz

Most spans cluster at short durations (tight yield loops); occasional high **y** points mark iterations that waited longer on the shared mutex or area semaphore. Compare ids **4–6** (shorter per-iteration work) vs **1–3** (heavier stress tests) in the same table.

Pairing algorithm At parse time, the viewer collects all `interval_start` / `interval_stop` STI events, sorts them by time (start before stop at the same timestamp), and pairs them with a **per-key stack** (LIFO).

Note parsing (last BTF CSV field):

Note format	Pairing key	Timeline / stats row id
`{id} tid:{task_id}` (current firmware)	interval `id` + `task_id`	`id` only (`Interval N`)

Note format	Pairing key	Timeline / stats row id
`{id}` or other legacy text	full note string	same as note

When `tid:` is present, concurrent workers can share the same interval **id** without cross-pairing: task 7's `START(1)` only pairs with task 7's `STOP(1)`, even if task 8 uses id 1 at the same time. Legacy traces without `tid` keep the original behaviour (pair by note / id string only).

1. **`interval_start`** — push the event onto that key's stack.
2. **`interval_stop`** — pop the most recent unmatched start for that key and form one instance `[start_time, stop_time]`.
3. **`interval_stop` with an empty stack** — ignored (orphan stop).
4. **Unmatched starts after the trace ends** — counted internally but not shown in statistics or on the timeline.

An **Interval N** row appears only when at least one **start→stop** pair exists for that id. If firmware emits `interval_start` without matching `interval_stop` events (same id and `tid` when present), that id is omitted entirely.

Example — interval id with no paired spans:

Interval id	`interval_start`	`interval_stop`	Viewer
0	50	50	**Interval 0**
1	100	**0**	*(no row — nothing to pair)*
2	50	50	**Interval 2**

If firmware emits `interval_start` for an id but never records matching `interval_stop` events (same id and `tid` when present), that id is omitted from the timeline and statistics. Ensure every `traceINTERVAL_START(id)` has a corresponding `traceINTERVAL_STOP(id)` with the same task id in the note when `tid:` is used.

This is **LIFO (last-in, first-out) nesting** within each pairing key.

```
# Nested on one task (same pairing key)
START(id=2)           stack: [A]
  START(id=2)         stack: [A, B]
  STOP(id=2)    →     pairs B→B  (inner span)
STOP(id=2)      →     pairs A→A  (outer span)

# Two tasks, same interval id, with tid in note
Task 7: START  note=1 tid:7
Task 8: START  note=1 tid:8
Task 7: STOP   note=1 tid:7   → pairs with task 7 start only
Task 8: STOP   note=1 tid:8   → pairs with task 8 start only
```

Result: **two** instances — one for the inner region, one for the outer region — each with the correct duration. **Min / Avg / Max / p95** in the statistics table are computed over **all** paired instances whose time range overlaps the active scope (full trace or cursor range). Timeline display uses a separate rule (see below).

Limitations **1. Concurrent overlap with the same id (cross-pairing)** — *legacy traces without `tid`*

When the BTF note has **no** `tid:{task_id}` suffix, pairing uses the note string only. The stack algorithm then assumes stops arrive in the **reverse order** of starts. That holds for true call-stack nesting on one thread, but **not** when several tasks use the **same interval id** at the same time.

```
Task on Core_1: START(2) ──────────────────────── STOP(2)
Task on Core_2:      START(2) ──────────── STOP(2)
Event order:       S1          S2         S1      S2
                   └─ stack pairs S2 with S1's STOP (wrong)
```

When starts and stops interleave across cores, the viewer can pair one task's **start** with another task's **stop**. That produces:

- **Count** — still the number of stop events that found a start on the stack (often equals the number of loop iterations).
- **Min / Avg / Max / p95** — can be **wrong**: bogus **very long** spans inflate max and avg; the distribution chart shows outlier points far above the real iteration time.

`example-4cores.btf.gz` is recorded **with** `tid` in each interval note, so ids **1–4** pair correctly across parallel workers. **Legacy** traces that omit `tid` may still cross-pair when several tasks share one id:

Interval id	Test (see `Demo/examples/freertos_test/main.c`)	Without `tid` — typical symptom
1	Context-switch / mutex stress (`vCtxSwitchWorker`)	Inflated **Max** / outlier scatter points
2	Mutex workers (`vMutexWorker`)	Same
3	Area-semaphore workers	Same
4	Nested interval stress	Same

For legacy traces, treat **short-duration clusters** in the scatter/histogram as the meaningful iteration times; treat isolated **very large** points and the table **Max** as pairing artefacts unless you have verified LIFO ordering for that id.

2. SMP task migration — why pairing is not done by core

The viewer pairs by **interval id** (and **task id** when `tid` is present), not by the **core** field on each STI event. A core-based matcher might seem attractive when several tasks share one id, but on **SMP** it is **not reliable**:

- A single FreeRTOS task can **migrate** between cores while an interval is open (START on Core_0, body runs, STOP on Core_2 after migration).
- Preemption and scheduling can also make the “running core” at START/STOP differ from the core where most of the work ran.
- Matching START and STOP by core would then **fail to pair** valid spans or would **split one logical interval** into orphan events.

```
Task A (may migrate):
  START(2) on Core_0  … runs on Core_0, then Core_1 …  STOP(2) on Core_1
  Core-based match: no single core owns both ends → broken or missed pair
  Id + tid stack: pairs correctly if no other task uses id 2 concurrently
```

So the viewer does **not** use core as a pairing key. With **`tid` in the note**, several tasks may share the same interval id safely; without `tid`, prefer **one interval id per logical scope** or accept LIFO-only pairing. Core columns on paired instances (`start_core` / `stop_core` in the parsed data) are informational only.

3. True time nesting vs concurrent overlap

Situation	Pairing correct?	Statistics meaningful?
Nested START/ST0P on **one thread** (LIFO order)	Yes	Yes — each nesting level is a separate instance
Concurrent tasks, **different** ids	Yes	Yes — ids are independent
Concurrent tasks, **same** id, **with `tid` in note**	Yes	Yes — per-task pairing keys
Concurrent tasks, **same** id, **no `tid`** (legacy)	Often **no**	**Count** ok; min/avg/max may be skewed
Start without matching stop (crash / trace cut-off)	Partial	Unmatched starts excluded from stats

4. Timeline bars vs statistics count

The statistics table and distribution charts use **every** paired instance in scope. The timeline draws **top-level spans only**: if instance B's `[start, stop]` lies entirely inside instance A's range (same id), only A is drawn. This is a **display** rule only; it does not change **Count** or min/avg/max.

View	What you see
Statistics → Count	All paired spans (including nested and cross-paired)
Distribution chart	One point per paired span (same set as Count)
Timeline → Interval N row	Non-nested spans only (fully contained children hidden)

Typical timeline bar counts for the full `example-4cores.btf.gz` trace:

Interval id	Statistics **Count**	Timeline bars (top-level)
1	480	5
2	480	7
3	240	1
4	480	1

Interval 2 on the timeline: **7** bars (one long outer span plus six shorter top-level spans), while the table still reports **Count = 480**. Use the distribution chart (click a scatter point) or a narrow cursor range to jump to one specific paired instance.

5. Instrumentation guidance

To get reliable per-task interval statistics:

- **Preferred (firmware in this repo):** use `traceINTERVAL_START(id)` / `traceINTERVAL_STOP(id)` — the logger records **`tid:{task_id}`** in the BTF note so parallel workers can share the same numeric `id`.
- **Legacy traces** without `tid`: use a **distinct interval id per task** (or per logical scope), not one shared id across parallel workers.
- For nested regions on a single thread, reuse the same id — LIFO pairing matches START/STOP nesting within each pairing key.
- Do **not** rely on **core** to disambiguate pairs on SMP: tasks can **migrate** between START and STOP (see limitation 2 above).
- Orphan stops (stop without start) are dropped; orphan starts at end of trace are not counted in the table.

Timeline rendering Interval bars are drawn as **solid** spans in the interval's colour. **Start** and **stop** events are marked with vertical tick lines (solid at start, dashed at stop) on the interval row.

Trace Health (TICK) Uses STI **TICK** timestamps to estimate scheduler tick regularity and detect whether the trace was recorded with the standard periodic tick or FreeRTOS **tickless idle** (`configUSE_TICKLESS_IDLE`).

Formula — for consecutive TICK events at times *t*n:

$$\Delta_n = t_n - t_{n-1}, \quad \mu = \text{mean}(\Delta_n), \quad \sigma = \text{stdev}(\Delta_n), \quad \text{CV} = \frac{\sigma}{\mu}$$

Missed ticks (est.) counts large gaps where Δn ≫ μ (roughly ⌊Δn / μ⌋ − 1).

What it tells you: In **TICK** mode (CV ≤ 5 %), intervals cluster at one nominal period — the scheduler clock is steady. **TICKLESS** mode (CV > 5 %) widens the distribution because idle periods suppress the tick interrupt; tall spikes in the scatter plot are multi-tick sleeps, not necessarily overload. Large **Max gap** with **good** status may still be worth inspecting for long critical sections or trace dropouts.

Field	Meaning
Status	`good` / `warning` / `critical` based on gap threshold
Mode badge	TICK (blue) or TICKLESS (amber) — detected automatically from the coefficient of variation (CV = σ/μ) of consecutive tick intervals. CV > 5 % is classified as tickless. Hover the badge to see the exact CV value
Ticks	TICK event count in scope
Avg period / Max gap	Observed tick spacing
Missed ticks (est.)	Rough count of skipped ticks from large gaps

In **tick mode** the timer interrupt fires at a constant rate and tick intervals form a tight cluster. In **tickless mode** the scheduler suppresses the tick interrupt during idle periods to save power, so consecutive intervals span one or many nominal tick periods — the distribution widens significantly.

When tick intervals can be charted, a **Tick Distribution…** button (bar-chart icon, theme-aware amber/orange styling) appears beside the **TICK** / **TICKLESS** mode badge when **≥ 2 ticks** are in scope (Desktop + Web). Clicking it opens the standard scatter + histogram popup showing: - **Scatter plot** — each tick interval over trace time; long idle periods appear as tall spikes. - **Histogram** — interval distribution with the same adaptive scaling and CDF overlay as other metric charts (auto may

choose p5–p95 or log duration when tickless idle stretches the range); clearly multi-modal in tickless mode (one sharp peak at 1 × period, another at 2×, 3×, etc.). The CDF helps quantify what fraction of tick intervals are a single period vs two or more.

Click any scatter point to jump to that tick time, add an **annotation**, and open the **Marks** tab with the annotation selected (same behaviour as other metric distribution charts).

Tick Distribution in `example-8cores.btf.gz` (~2496 TICK events, CV ≈ 35.9 % → **TICKLESS**):

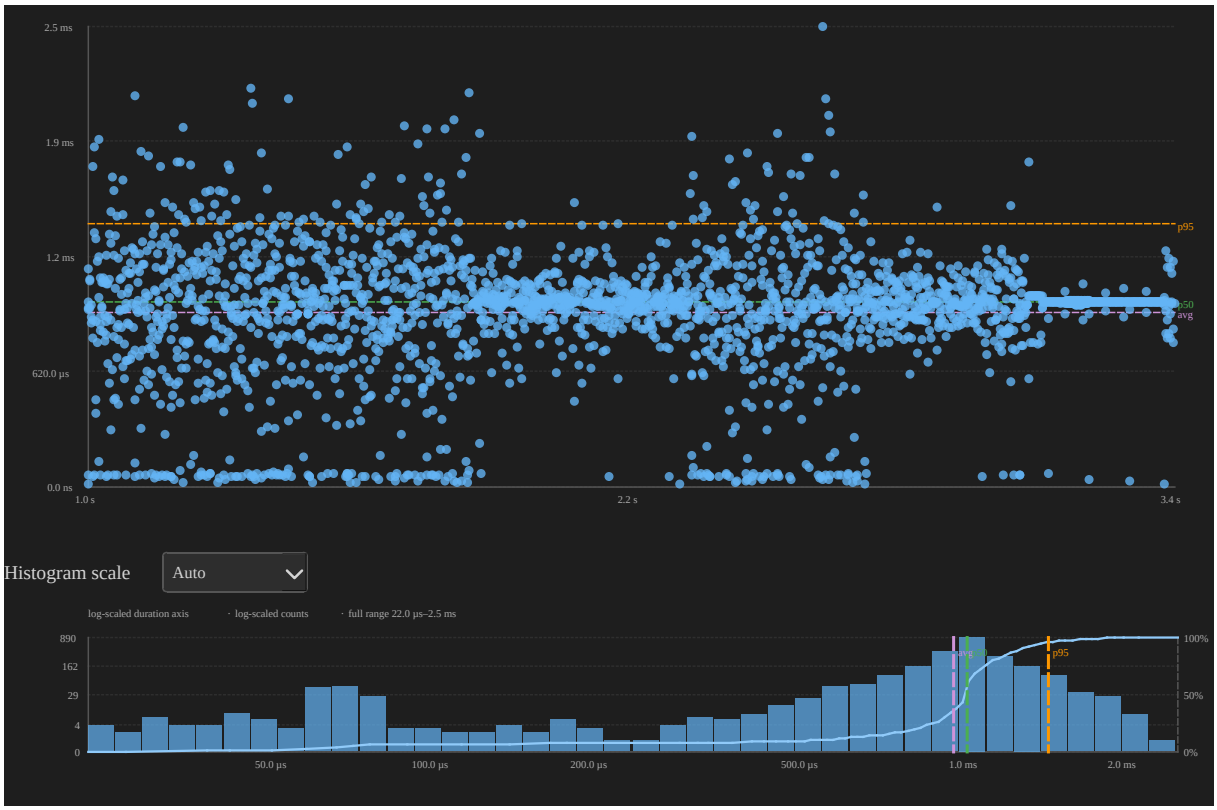


Figure 11: Tick interval distribution chart — scatter and histogram of consecutive TICK gaps in example-8cores.btf.gz

The histogram's multiple peaks (1×, 2×, 3× the nominal period) confirm tickless idle: most gaps are a single tick, but idle stretches skip several nominal periods before the next TICK fires.

Large gaps may indicate CPU overload, long critical sections, tickless idle, or tracing gaps — not necessarily a FreeRTOS configuration error.

Tag Analysis Aggregates numeric samples from the 8 general-purpose STI **tag** channels (`tag0_event` … `tag7_event`, plus the unindexed `tag_event`) — free-form values emitted by firmware via `btf_traceTAG(id, value)` for any application-defined metric that doesn't fit an existing STI channel (queue depth, ADC reading, free heap, sensor reading, etc.). One row appears per channel that has at least one sample.

Formula — over all sample values *v*k on a channel in scope: **Count** is the number of samples; **Min** / **Avg** / **Max** are the usual statistics; **p95** is the 95th-percentile value.

What it tells you: Unlike the other metric tables, the tag y-axis is the **raw application value**, not a duration — so what it means depends entirely on what firmware puts in the channel. A widening spread or a rising trend in the scatter plot can flag a slow memory leak (free heap), growing backlog (queue depth), or drifting sensor reading; a tight cluster around **Avg** with occasional **p95**/**Max** outliers is normal sampling noise.

Column	Meaning
Channel	Display name (`Tag 0` … `Tag 7`, or `Tag` for the unindexed `tag_event`)
Count	Number of samples in scope
Min / Avg / Max / p95	Sample value statistics (not a time duration)

Distribution chart — click any row:

- **Scatter:** x = sample time, y = tag value.
- **Histogram:** distribution of tag values.

tag0_event in `example-8cores.btf.gz` (2330 samples, min 8,144, avg ≈ 36,845, max 71,904, p95 41,936):

Figure 12: Tag value distribution chart for tag0_event in example-8cores.btf.gz

Shown only when the trace contains `tag0_event` … `tag7_event` (or `tag_event`) STI samples.

Core migration analysis

A **migration** is recorded when consecutive slices of the same task (merge-key) run on different cores. Migrations are detected at parse time from the segment timeline — there are no separate markers drawn on the timeline; use the **Core Migrations** table, **Migration heatmap**, **Migration chord diagram**, **Trace Compare…**, or Find **Migrations** mode to inspect them.

Highlight a migrating task on the timeline To *see* a task hop cores (not only read rates in a table):

1. Switch to **Core View** (toolbar **Core**).

2. Type the task name in the legend **filter** (e.g. `CS[22]`) so only that task's sub-rows remain.
3. ⊞ **Expand All** (or expand each core the task visited).
4. Click the task label (or a segment) to **lock-highlight** it — other tasks dim; the highlighted bars jump from core row to core row at each migration.
5. Optionally place cursors around the burst and enable **Limit to cursor range** so **Statistics → Core Migrations** and the heatmap recompute for that window only.

Headless equivalent (`example-8cores.btf.gz`, CS stress burst where `CS[22]` visits all 8 cores):

```
python builds/btf_viewer.py snapshot ../tracedata/example-8cores.btf.gz \
    -o ../images/stats/tasks-migrate-cs22.svg \
    --view timeline --view-mode core --task "CS[22]" \
    --lo 1805000 --hi 1865000
```

Or regenerate with `make -C BTFViewer update-images`.

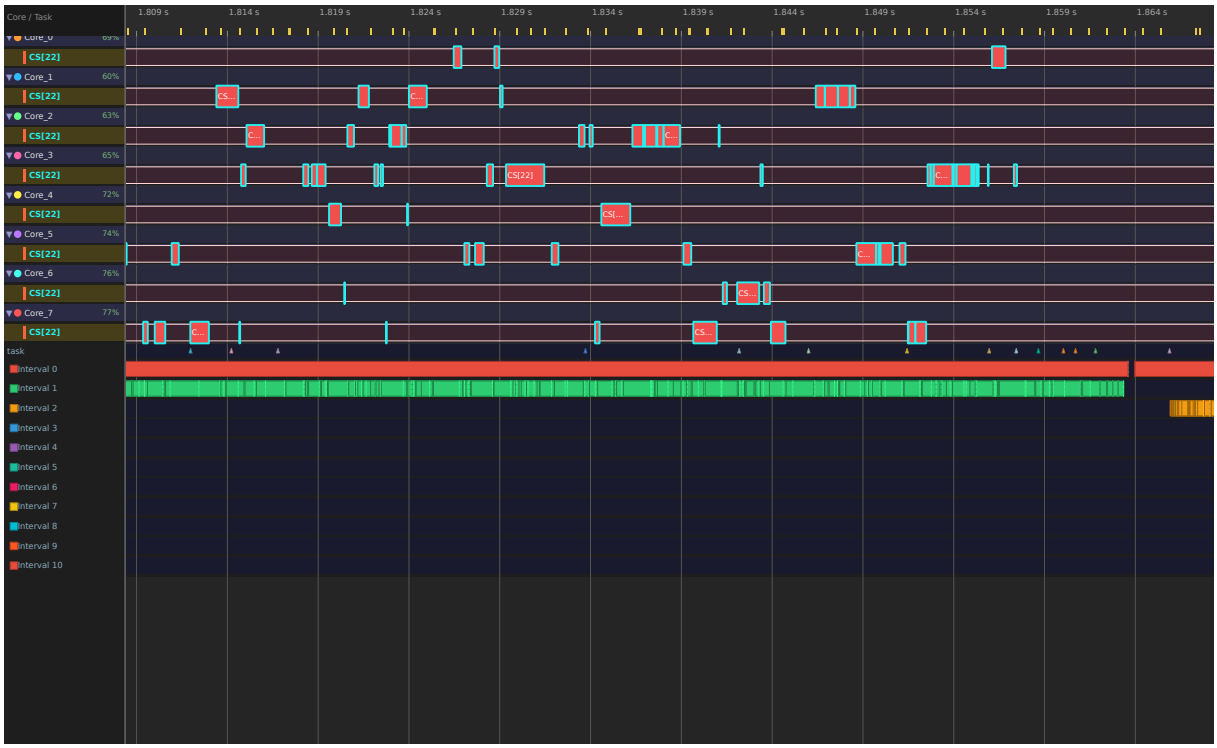


Figure 13: Core View: CS[22] highlighted hopping across Core_0…Core_7 (example-8cores.btf.gz)

~60 ms window (`--lo 1805000 --hi 1865000`). Cyan outline = locked highlight; each core row shows only `CS[22]` after the task filter — gaps between bars are off-CPU / other cores.

Inspect migrations involving a specific core After highlighting a hot task:

1. **Statistics → Core Migrations** — note **Primary** core, **Rate**, **Dwell**, **Ping** for that task. Click the row → **Dwell** / **Rate** / **Gap** charts.
2. **Core-Pair Migration Summary** — sort or scan pairs where that core is **From** or **To** (e.g. `Core_5→Core_7`) to see which neighbour absorbs the traffic and whether **Bounce %** is elevated.
3. Toolbar **Heatmap** — find the dark cells on rows that include the core of interest; click to drill to the per-task grid, then click the task cell to zoom/filter the timeline to that pair/window.
4. Toolbar **Chord** — hover the core's arc to highlight only its ingress/egress chords.
5. **Core Time Breakdown** — click a core row to jump Core View focused on that core.
6. Headless CSV for a scoped window:

```
python builds/btf_viewer.py migrations ../tracedata/example-8cores.btf.gz \
    --lo 1805000 --hi 1865000 -o - | head
```

Legend panel: check **Migrated tasks only** to hide tasks that never left their first core.

Statistics → Core Migrations (collapsible section) — tasks that ran on two or more cores:

Migration rate — normalizes raw migration count against task active time and (when TICK STIs exist) scheduler ticks, so a task that migrates often relative to how much it runs stands out:

$$R_m = \frac{N_{\text{migrations},i}}{T_{\text{exec},i}}$$

The **Rate** column shows *R*m as migrations per second of on-CPU time (e.g. `1.23/s`). When the trace includes TICK events, it also shows migrations per **on-CPU** scheduler

tick for that task (e.g. `2.785/tick`) — TICK STIs that fall inside one of the task's slices in scope, not trace-wide tick count.

What it tells you: A high rate means the task is **bouncing** between cores (thrashing). For high-priority real-time tasks you ideally want this close to zero.

Average core dwell time — mean duration of each on-CPU stay before the task blocks, yields, or migrates:

$$\bar{T}_d = \frac{1}{N_{\text{slices}}} \sum_k d_k = \frac{T_{\text{exec},i}}{N_{\text{slices},i}}$$

Each slice *d*k is one switch-in episode (equivalent to averaging per-core dwell, *T*on / *N*slices, on each core the task visited).

What it tells you: If **Dwell** is extremely short (e.g. less than a few milliseconds or close to your system tick period), the scheduler is spending disproportionate effort moving the task between cores instead of letting it compute.

Column	Meaning
Task	Display name (`Name[id]`)
Migr	Migration count in the current scope (full trace, or cursor range when **Limit to cursor range** is on)
Rate	Migration rate — `/s` of task active time; `/tick` = migrations per on-CPU TICK for this task in scope
Dwell	Average on-CPU slice duration (core dwell time) in scope
Cores	Distinct cores with on-CPU time or migrations in the current scope
Primary	Core with the most active time in scope, with its share (%)
Ping	Ping-pong migrations — three consecutive migrations A→B→A within 1 µs

Column	Meaning
STI±	Migrations with an STI event within ±500 ns
Gap after	Average off-CPU gap immediately after a migration
Gap other	Average blocking gap elsewhere for the same task

Click a row to open a **distribution chart** with three in-dialog tabs (switches the chart in place, no need to close and reopen):

- **Dwell** (default tab) — one point per on-core run: x = run start time, y = run duration d_k (same definition as **Average core dwell time** above, but per-sample instead of averaged).
- **Rate** — one point per migration after the first: x = migration time, y = time since the task's *previous* migration. Clusters of short gaps mean bursts of rapid bouncing; a flat, evenly spaced scatter means migrations are rare and spread out.
- **Gap** — one point per migration with a positive **Gap after** value: x = migration time, y = off-CPU gap immediately following that migration (the same samples averaged into the **Gap after** column; **Gap other** is not plotted here — open the **Blocking Time** chart for the same task instead, since it covers all off-CPU gaps).

Each tab's scatter + histogram uses the same **adaptive scaling** as other metrics charts (see Metrics Distribution Charts).

CS[22] in `example-8cores.btf.gz` (a context-switch stress task that migrates often):

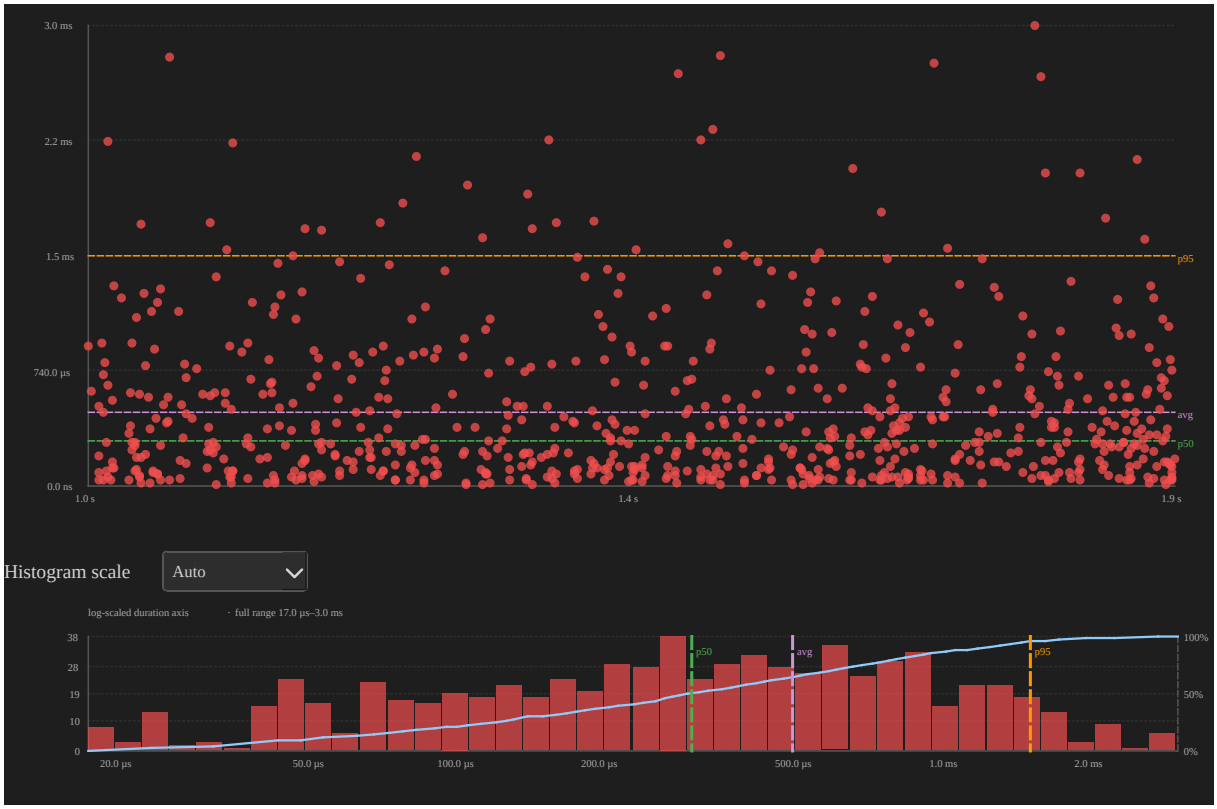


Figure 14: On-core dwell time distribution for CS[22] in example-8cores.btf.gz

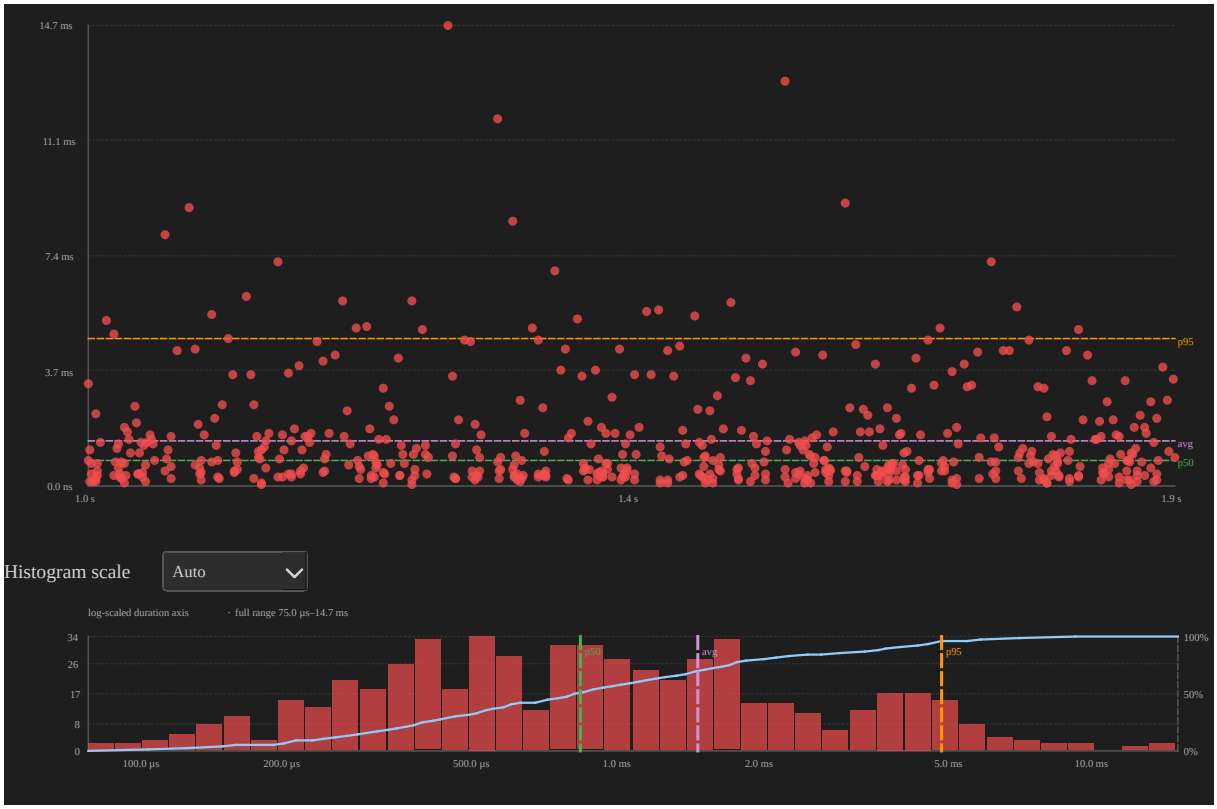


Figure 15: Time between migrations for CS[22] in example-8cores.btf.gz

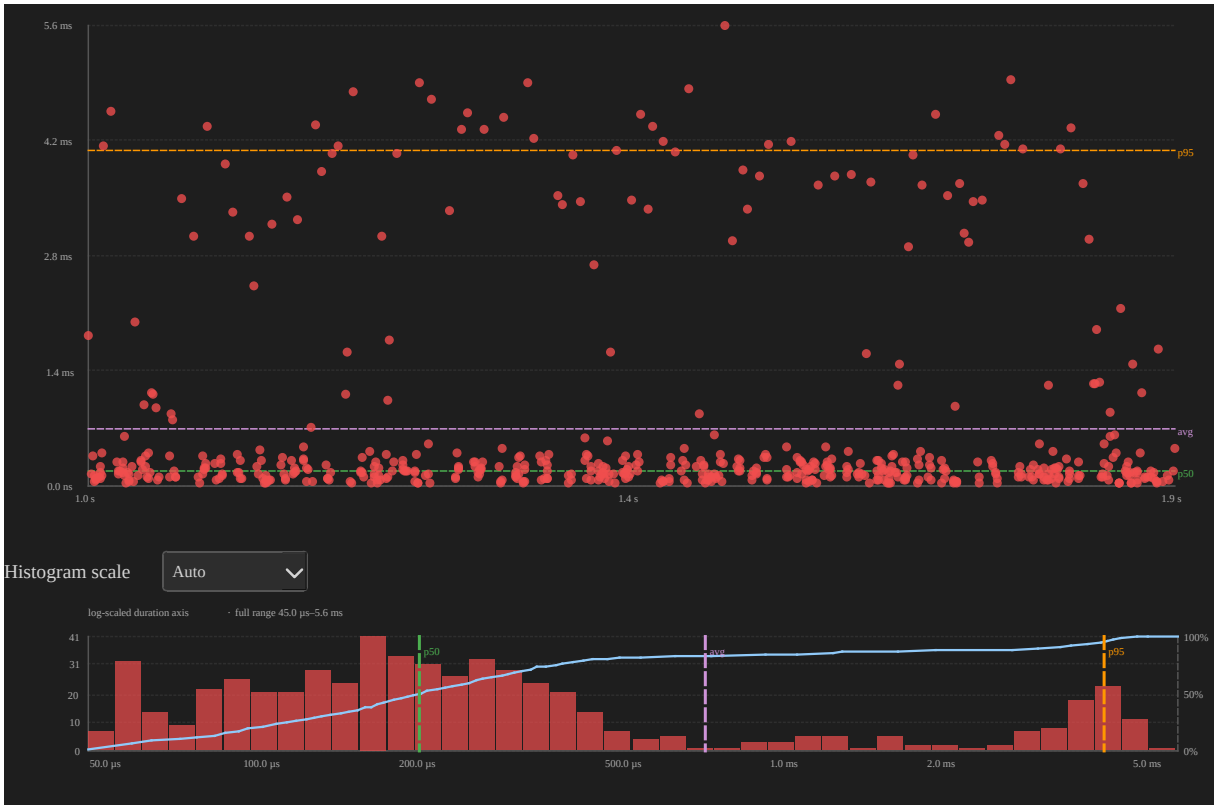


Figure 16: Post-migration gap distribution for CS[22] in example-8cores.btf.gz

Drag the resize handle below the table to show more or fewer rows.

Migration heatmap Visualise **when** migrations happen between core pairs — complementary to the per-task **Core Migrations** table. Traces with **more than 16 cores** use a **three-level** drill-down (core×core matrix → outgoing pairs × time bins → tasks). Smaller multi-core traces use a **two-level** flow (core-pair rows × time bins → tasks).

Bounce-only filter — the **Show: All Migrations / Show: Bounce Only** toggle button in the heatmap dialog restricts the visualisation to migrations that occurred while a mutex was held across two different cores (lock-bounce migrations, tracked via `lock_bounce_migration_ns`). Use this to identify which core pairs and time windows are dominated by cache-line bouncing rather than normal scheduling migrations.

Example (`example-4cores.btf.gz`) — screenshots exported with **Export SVG** from the heatmap dialog (full current drill level, no hover highlight):

Level 1 — core-pair overview (4 cores · 12 time bins across the trace):

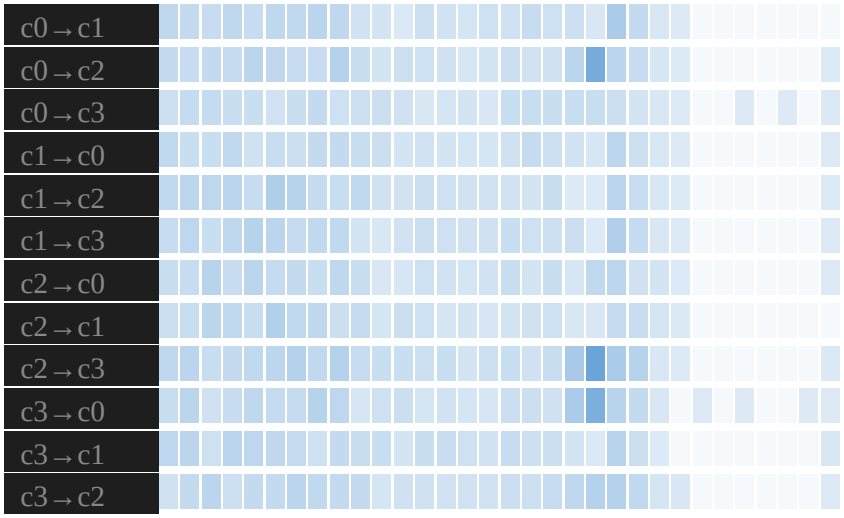


Figure 17: Migration heatmap Level 1: core-pair rows and time bins for example-8cores.btf.gz

Each row is a directed core pair (`c0→c1`, `c0→c2`, …). Cell colour intensity is the migration count in that bin (darker blue = more events). Horizontal bands show **when** traffic occurred — e.g. repeated activity on `c0→c2` and `c1→c0` during the context-switch stress phases. Click a non-empty cell to drill into tasks for that pair and time window.

Level 2 — task grid (after clicking a cell on the pair overview):

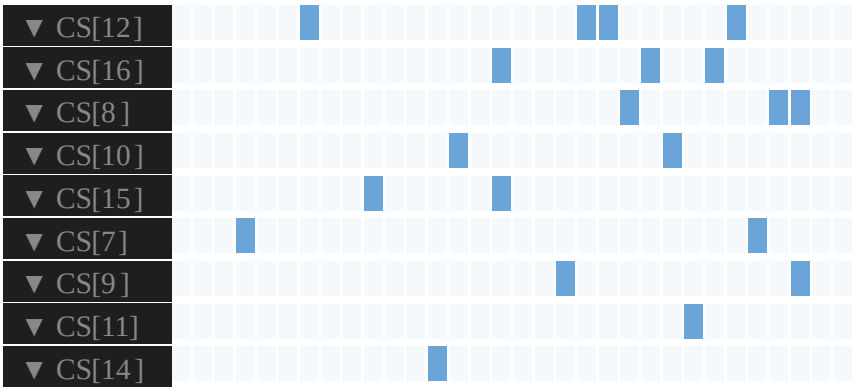


Figure 18: Migration heatmap Level 2: per-task sub-bins after drilling from example-8cores.btf.gz

Rows are tasks that migrated on the selected core pair within the chosen bin; columns are **12 sub-bins** spanning that bin's time window. Brighter cells mark sub-intervals where that task crossed cores most often. Each row label is prefixed with a directional indicator: ▲ = task primarily migrates in this direction (egress-dominant), ▼ = more migrations on the reverse path (ingress-dominant), ⇄ = roughly symmetric. Click a task cell to zoom the timeline, place **C1** / **C2** at the sub-bin edges, switch to **Task View**, and filter to that task only.

Typical workflow (≤ 16 cores)

1. Open **Heatmap** from the toolbar (`tracedata/example-8cores.btf.gz` or `example-16cores.btf.gz` are good demos).
2. **Level 1** — click a hot cell on a core-pair row to open the task grid for that bin.
3. **Level 2** — click a task cell to zoom the timeline, place **C1** / **C2**, switch to **Task View**, and show only that task.
4. **Show all tasks** — when done, clear the task filter and restore the timeline viewport, cursors, and highlights from before the heatmap was opened; the heatmap returns to the core-pair overview.

Typical workflow (> 16 cores)

1. **Level 1 — core×core matrix** — rows = source cores, columns = destination cores. Hover a row to highlight it; **click a row** to drill into outgoing migrations from that core.

2. **Level 2 — outgoing pairs** — one row per destination (`c3→c1`, `c3→c2`, …), columns = 32 time bins. Hover highlights the row; click a cell to open the task grid for that pair and bin.
3. **Level 3 — tasks** — same as Level 2 above for smaller traces.
4. **← Back** steps up one level; **Show all tasks** resets the timeline filter and returns to Level 1 (matrix or pair overview).

Open	Toolbar **Heatmap** (Desktop + Web). Enabled only when the trace has **2 or more cores** (single-core traces disable the button). Desktop: non-modal dialog (timeline stays interactive). Web: semi-transparent overlay.
Level 1 (≤ 16 cores)	One row per directed core pair (`c0→c1`, `c1→c0`, …). Core names are shortened in the labels (e.g. `Core_0` → `c0`).
Level 1 (> 16 cores)	**Core×core matrix** — row = source core, column = destination core. Hover highlights the row; **click a row** (not a single cell) to open outgoing pairs.
Level 2 (> 16 cores)	Outgoing pairs from the selected source core (`c3→c1`, `c3→c2`, …) × 32 time bins. Row hover highlight; click a cell for tasks.
Level 1 columns	**32 equal time bins** across the scoped window. Cell colour intensity is the migration count in that bin (darker = more migrations). Hover for time range, count, and task names.
Level 2 rows	Task display names that migrated on the selected core pair in the selected bin.
Level 2 columns	**32 sub-bins** within the Level 1 cell’s time window.

← Back	Step up one level (tasks → outgoing pairs → matrix, or tasks → pair overview on smaller traces).
Click cell	≤ 16 cores: Level 1 → tasks. > 16 cores: Level 2 → tasks. Details below.
Click row	> 16 cores only: matrix Level 1 → outgoing pairs for that source core.
All tasks (toolbar)	Shown next to **Heatmap** while a heatmap task filter is active. Same reset as **Show all tasks** below.
Show all tasks	Clears the task filter and **restores the timeline state captured when the heatmap was opened** (viewport, cursors, highlights, view mode). The heatmap returns to **Level 1**; heatmap time scope follows the restored cursors (full trace if fewer than two cursors were saved). Available from: toolbar **All tasks**, heatmap dialog **Show all tasks**, Legend **Clear** (Web: Marks page; Desktop: legend dock), or enabling **Migrated tasks only**. Loading a new trace or switching tabs also clears the filter (Desktop closes the heatmap dialog on tab switch).
Export PNG / SVG	Footer buttons export the **full current drill level** (all rows/columns, no hover highlight). Filenames: `heatmap-{level}-{timestamp}.png` or `.svg` (e.g. `pairs`, `matrix`, `outgoing`, `tasks`). Desktop: file save dialog + status bar confirmation. Web: direct download.

Scope	**Full trace** by default. With **2 or more cursors** placed, Level 1 uses the time window from **C1** through the last cursor (subtitle shows the range). Independent of the Statistics panel **Limit to cursor range** toggle. **Show all tasks** restores pre-heatmap cursors, so heatmap scope matches that saved window.
Empty state	*No migrations in scope.* (Level 1) or *No task migrations in this cell.* (Level 2) when no events fall in the current window.

What happens when you click a cell Only cells with at least one migration (count > 0) are clickable.

≤ 16 cores — core-pair overview (rows = `c0→c1`, … · columns = 32 time bins)

1. **Click a cell** — Opens the task grid for that core pair and time bin.

> 16 cores — core×core matrix (rows/columns = cores)

1. **Hover a row** — Highlights the source core row.
2. **Click a row** — Opens outgoing pair rows (`cN→c1`, `cN→c2`, …) × 32 time bins.

Outgoing pairs / ≤ 16 cores Level 1 (rows = core pairs · columns = 32 time bins)

3. **Click a cell** — Opens the task grid for that pair and time bin.

Task detail (rows = task names · columns = 32 sub-bins)

4. **Click a cell** — The timeline zooms to that sub-bin, **C1** and **C2** are placed at the sub-bin edges, the view switches to **Task View**, and only that **task** row is shown.
5. **Another task cell** — Each click **replaces** the previous filter (last task wins).

Reset — Show all tasks

6. Clears the task filter, restores the viewport/cursors/highlights saved when the heatmap was opened, returns the heatmap to **Level 1** (matrix or pair overview), and shows every task row again.

Use the heatmap to spot bursts of cross-core traffic, drill into the contributing tasks, then jump to Task View for slice-level inspection. For aggregate per-task migration statistics (ping-pong, STI correlation, gap-after), use **Core Migrations** in the Statistics panel.

Migration chord diagram Visualise **how much** migration traffic flows between each pair of cores, as directional chords around a circle — one arc per core. Complementary to the **Migration heatmap** (which shows *when* migrations happen): the chord diagram gives a single, at-a-glance summary of *volume and direction* across the whole scoped window, with no drill-down levels.

- **Chords** — a curved band between two core arcs for each directed pair with at least one migration; chord width is proportional to migration count and its colour fades from the source core's colour to the destination core's colour. Pairs that migrate in both directions are drawn as two offset chords.
- **Hover a core arc** — highlights that core's arcs and fades all other chords/arcs; the fixed-height line below the diagram shows a breakdown (`out` / `in` totals, then each non-zero directed pair count for that core).
- **Bounce-only filter** — the same **Show: All Migrations / Show: Bounce Only** toggle as the heatmap, restricting chords to lock-bounce migrations (held across a core boundary).
- **Scope** — same as the heatmap: full trace by default, or **C1** through the last cursor when 2 or more cursors are placed.

Open	Toolbar **Chord** button (Desktop + Web). Enabled only when the trace has **2 or more cores**. Desktop: non-modal dialog (timeline stays interactive; closes if you switch to a different trace tab). Web: semi-transparent overlay (closes on tab switch).

Export PNG	Footer button exports the diagram exactly as rendered (no hover highlight). Filename: `migration-chord-{timestamp}.png`. Desktop: file save dialog. Web: direct download.
Export SVG	Footer button next to Export PNG; same content, vector format. Filename: `migration-chord-{timestamp}.svg`. Desktop: file save dialog. Web: direct download.
Empty state	*No migrations in scope.* when no migrations fall in the current window.

Example (`example-8cores.btf.gz`) — screenshot exported with **Export SVG** from the chord dialog (full trace scope, no hover highlight):

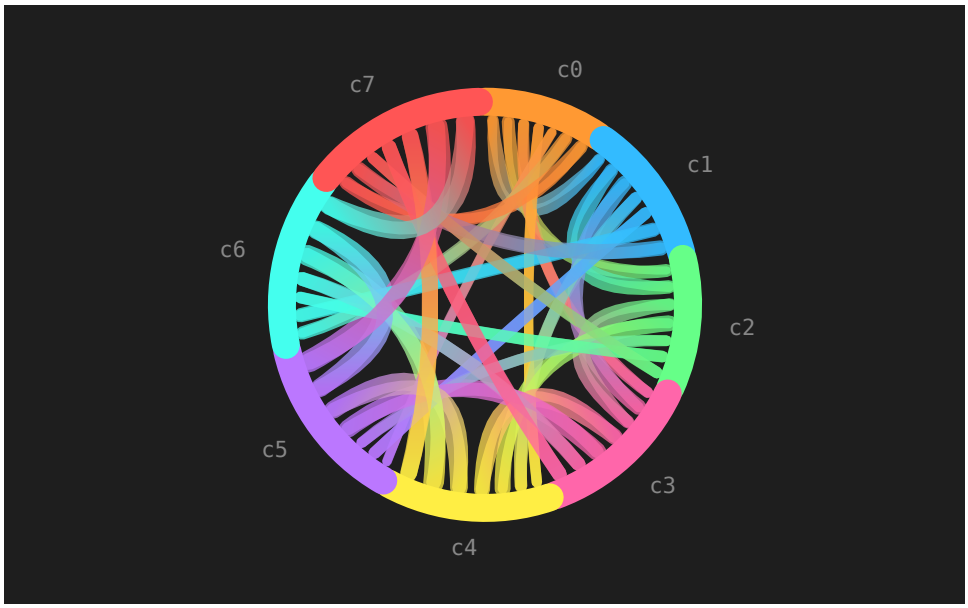


Figure 19: Migration chord diagram: directional core-to-core migration volume for example-8cores.btf.gz

Each arc around the circle is one core (`c0`…`c7`); a chord between two arcs is a di-

rected migration flow, coloured as a gradient from its source core to its destination core, with width proportional to migration count. Thicker chords between adjacent-looking core pairs (e.g. `c0↔c2`, `c1↔c6`) stand out immediately as the busiest migration paths, without needing to drill through a heatmap grid first.

Trace Compare… Compare two traces you already have open side-by-side:

1. Open at least **two** `.btf` files (Desktop: **File → Open** adds a tab; Web: **Open** adds a tab in the bar under the toolbar).
2. In the **Statistics** panel footer, click **Trace Compare…** (enabled when two or more tabs are loaded).
3. Choose **Trace A** and **Trace B** from the dropdowns.
4. Optionally check **Limit to each tab's cursor range** to compare metrics within C1–Cn on each trace (requires 2+ cursors per tab).
5. Switch between **Summary**, **Top Tasks**, and **Core Migrations** tabs.

By default, compare views use the **full trace**. With the cursor-range checkbox enabled, each side uses that tab's own cursor window independently.

Summary — high-level diff:

Metric	Notes
Span	Total trace duration (or cursor-range width when scoped)
Tasks / Segments / STI events	Counts
Context switches	Total across all cores
Core gap avg / max	Idle time between consecutive slices on each core
Migrations (total) / Migrated tasks	Core-migration counts

Each row shows Trace A, Trace B, and **Δ** (signed difference).

Top Tasks — top 10 user tasks by CPU% from each trace, unioned by display name (`Name[id]`). Tasks present in only one trace show **—** for the missing side.

Core Migrations — same columns as the Statistics panel, compared side-by-side:

Column	Meaning
Task	Display name (`Name[id]`)
Migr A / **B**	Migration count in scope for that trace
Δ	Difference (A − B)
Rate A / **B**	Migration rate label (`/s` and `/tick`) per trace
Rate Δ	Signed difference of migrations per second of on-CPU time (A − B)
Dwell A / **B**	Average on-CPU slice duration per trace
Dwell Δ	Signed difference of average dwell time (A − B)
Ping A / **B**	Ping-pong count in each trace

Use this to compare builds, configurations, or runs of the same workload without merging traces manually.

Find → Migrations: lists migration boundary times; F3 / `Shift+F3` jump between them (Desktop + Web).

Find & Jump

The **Find** tab in the right-side panel (**Statistics** / **Marks** / **Find**) is shared by Desktop and Web. Open it via the toolbar **Find** button, **Navigate → Find** (`Ctrl+F`), or by selecting the **Find** tab.

Action	Effect
Type in the search box	Highlight all matching positions on the timeline
`Enter` / F3 / **Navigate → Find Next**	Jump to the next match

Action	Effect
`Shift+F3` / **Navigate → Find Previous**	Jump to the previous match
Clear the search box	Remove search highlights

The status label shows the total number of matches and the current position (e.g. `12 matches (at 4)`).

Find modes

Select the search mode from the dropdown next to the search box:

Mode	Searches
Contains	Task names, annotation text, and STI notes that contain the query string
Exact	Exact full-string match on task names and annotations
Regex	Regular expression match across task names, annotations, and STI notes
Migrations	Core-migration boundary timestamps (query ignored; jumps to every migration event)
STI	STI events whose channel name or note contains the query
Intervals	`interval_start` / `interval_stop` events matching the query (channel or note)
Lifecycle	Task `create` / `delete` / `suspend` / `resume` events on the `task` STI channel
Pointers	Sync object pointer values (`0x…`) in STI notes (mutex / semaphore / queue)

Bookmarks & Annotations

Both are stored per-trace in `btf_viewer.rc` and restored automatically the next time the same file is opened. **Desktop:** bookmarks and annotations are listed in the **Bookm.** and **Anno.** sub-tabs under **Marks**. **Web:** both appear in a single marks list in the **Marks** tab.

Bookmarks

A **bookmark** is a simple timestamped marker — a flag pinned to a specific moment in the trace. Use bookmarks when you want to quickly return to a position you already understand.

Adding a bookmark: - Right-click anywhere on the timeline and choose **Add Bookmark here**. - Or, via **Navigate → Add Bookmark** (`Ctrl`+B) to place one at the viewport centre.

Annotations

An **annotation** is a timestamped note — you supply a short text description that is stored alongside the timestamp. Use annotations when you want to record *why* a moment is interesting (observations, bug descriptions, review comments, etc.).

Adding an annotation: - Right-click anywhere on the timeline and choose **Add Annotation here**; a prompt asks for the note text. - Or, via **Navigate → Add Annotation** to annotate the viewport centre.

Bookmark vs. Annotation — when to use which

Scenario	Use
You found the `CAN_Rx` preemption that triggers a deadline miss and want to jump back to it quickly	**Bookmark** — quick marker, no text needed
You want to leave a note — *“first occurrence of 3 ms latency spike on Core_2, ticket #492”* — for yourself or a colleague	**Annotation** — the text note travels with the `.rc` file

Scenario	Use
Marking several candidate events to compare before deciding which matters	**Bookmarks** — fast to place, easy to jump between
Documenting a code review of a trace, explaining each anomaly	**Annotations** — each one carries the explanation inline
Sharing a trace with a colleague and highlighting the two key moments for them to look at	**Annotations** — the notes appear without any external document needed

Managing marks

In the **Marks** tab:

- **Desktop:** use **Bookm.** / **Anno.** sub-tabs; **View → Marks** shows the panel dock.
- **Web:** bookmarks and annotations share one list.
- **Double-click** a bookmark or annotation row to jump to its timestamp.
- **Delete** key (or the **Delete** button) removes the selected mark.
- Bookmark labels can be renamed **inline** by double-clicking the label text.
- **Session** / **Import Session** — export or import a portable JSON file with cursors, marks, viewport zoom/pan, view options, Find query, and pinned highlight. The format is shared between Desktop and Web so you can move analysis state between platforms (open the same trace first, then import).

Right-Click Context Menu

Right-clicking anywhere on the timeline opens a context menu:

Item	Effect
Place cursor here	Add a cursor at the clicked timestamp
Remove nearest cursor	Remove the cursor closest to the click position
Clear all cursors	Remove all cursors
Add Bookmark here	Add a bookmark at the clicked timestamp
Add Annotation here	Prompt for note text, then add an annotation at the clicked timestamp

The **Add Bookmark** and **Add Annotation** items are only shown when a trace is loaded.

Recent Files

File → Open Recent lists the 8 most recently opened `.btf` files. Clicking an entry opens it in a new tab (or switches to the existing tab). The list is stored in `btf_viewer.rc` and persists across launches.

Session persistence (`btf_viewer.rc`)

Settings, window layout, bookmarks, and multi-tab state are stored in `btf_viewer.rc` next to `builds/btf_viewer.py`.

Section / key	Purpose
`[files] open_tabs_json`	JSON array of open trace paths (tab order)
`[files] active_tab_index`	Which tab was focused on exit

Section / key	Purpose
`[files] last_file`	Path of the active tab (legacy; also used as fallback when `open_tabs_json` is empty)
`[tab_view] trace_<hash>`	Per-trace zoom level, fit mode, and cursor positions
`[trace_state] trace_<hash>`	Per-trace bookmarks and annotations
`[view] label_width`	Task/core label column width in px (60–600); also saved per window size in `dock_profile_label_width`
`[view] cpu_splitter_bottom_h` / `cpu_splitter_user_sized`	Timeline vs CPU load splitter height (Desktop)
`[stats] table_height_<section>`	Per-section metric table heights in Statistics (Desktop)
`[zoom]` / `[cursors]`	Zoom and cursors for the last active tab (legacy compatibility)

Keyboard Shortcuts

Shortcuts marked **(W)** are Web-only. All others work on both Desktop and Web.

File & Tabs

Key	Action
`Ctrl+O`	Open `.btf` file (new tab)
`Ctrl+W`	Close active tab

Key	Action
`Ctrl+Tab`	Next trace tab
`Ctrl+Shift+Tab`	Previous trace tab
`Ctrl+Q`	Quit (Desktop)

Edit

Key	Action
`Ctrl+Z`	Undo last cursor / mark change
`Ctrl+Y`	Redo

View / Zoom

Key	Action
`Ctrl++`	Zoom in
`Ctrl+-`	Zoom out
`Ctrl+0` / `F`	Fit entire trace to window
`Ctrl+R`	Zoom to cursor range (requires ≥ 2 cursors)
`Ctrl+,`	Open Settings
`G`	Toggle grid lines on/off
`I`	Toggle STI event rows on/off
`D`	Toggle dark / light theme
`Double-click segment`	Zoom to that segment; double-click again to restore (Desktop)
`Double-click label edge`	Auto-fit label column width
`Double-click ruler`	Fit timeline to trace **(W)**

Navigation

Key	Action
`Ctrl+G`	Jump to a specific time (dialog)
`Ctrl+Home`	Jump to trace start
`Ctrl+End`	Jump to trace end
← / →	Scroll along time axis (horizontal) or row axis
↑ / ↓	Scroll along row axis or time axis (vertical)
`Shift+←` / `Shift+→`	Jump to previous / next segment boundary (horizontal)
`Shift+↑` / `Shift+↓`	Jump to previous / next segment boundary (vertical)
`Tab`	Next segment / next highlighted-task segment
`Shift+Tab`	Previous segment / previous highlighted-task segment

Export & Snapshot

Key	Action
`Ctrl+S` / S **(W)**	Open snapshot editor (capture viewport for annotation)
`Ctrl+Shift+C`	Copy viewport to clipboard (no editor)
`Ctrl+Shift+S`	Save viewport as SVG
`Ctrl+Shift+E`	Export Perfetto (Chrome Trace JSON)

Cursors

Key	Action
`C`	Place cursor at viewport centre
`Shift+C`	Clear all cursors

Find

Key	Action
`Ctrl+F`	Open Find panel
`F3`	Find next match
`Shift+F3`	Find previous match

Marks

Key	Action
`Ctrl+B` / `B` / `M`	Add bookmark at current position
`Shift+B`	Clear all bookmarks
`Ctrl+Shift+B` / `A`	Add annotation at current position
`Shift+A`	Clear all annotations

Mouse / Trackpad

Action	Effect
Scroll wheel	Pan vertically (rows) in horizontal layout; pan time in vertical
Shift+scroll	Swap axes
Ctrl+scroll	Zoom in/out around pointer

Action	Effect
Two-finger pinch (macOS)	Zoom in/out
Left-drag (background)	Pan timeline
Left-drag ruler	Pan timeline **(W)**
Middle-drag	Draw a time-range selection band → zoom in on release
Left-click (timeline)	Place cursor
Shift+left-click	Snap cursor to nearest segment boundary
Left-click (near cursor)	Remove that cursor
Shift+right-click	Clear all cursors
Right-click (timeline)	Remove nearest cursor / context menu
Double-click (segment)	Zoom to that segment; double-click again restores previous zoom (Desktop)
Double-click (ruler)	Fit timeline to trace **(W)**
Double-click (label edge)	Auto-fit label column width
Left-drag (label edge)	Resize label column
Left-drag (cursor line)	Move cursor to new position
Left-drag (mark flag)	Move bookmark or annotation

Other

- Hover over any segment bar or STI marker for a detailed tooltip (task, core, start/end/duration, slice index on core, previous/next task on that core, gap before the slice).
- Toggle STI events, grid lines, and hover highlight from **Settings** (`Ctrl+,`) or with the `I`, `G` keyboard shortcuts.
- Drag and drop a `.btf` file onto the window to open it in a new tab.

- Open tabs, active tab, zoom level, and cursor positions are saved per trace in `btf_viewer.rc` and restored on the next launch.
- Press ? (Web) to open the interactive keyboard shortcut reference panel.

Reference

Synthetic trace generation, BTF file format reference, and source code map. For practical diagnosis workflows (first look, load balance, WCET, thrashing, deadlines, exports, Perfetto, ...) see **WORKFLOWS.md**.

Generating synthetic traces — `scripts/gen_trace.py`

`scripts/gen_trace.py` generates a synthetic FreeRTOS-style BTF trace file for testing or demo purposes. Task names are drawn from a realistic embedded-system pool (`CAN_Rx`, `Motor_L`, `PID_Speed`, ...). The scheduler simulation includes task priorities, IDLE time, TICK ISRs, generic STI tags, and firmware-style **interval** (`interval_start` / `interval_stop`) and **mutex** (`create` / `take` / `give`) STI scenarios. A fast inline **xorshift32 PRNG** is used internally for high-throughput event generation (≈ 0.22 s for 100 000 events on typical hardware).

Quick start

```
cd BTFViewer

# defaults: 4 cores, 100 tasks, 8 K events  →  freertos_4c_100t_8k_events.btf
python3 scripts/gen_trace.py

# 4 cores, 50 tasks, 500 K events
python3 scripts/gen_trace.py -c 4 -t 50 -e 500000 -o my_trace.btf

# 16 cores, 200 tasks, 2 M events, 500 Hz tick
python3 scripts/gen_trace.py -c 16 -t 200 -e 2000000 --tick-hz 500

# Disable all STI; pin every task to one core
python3 scripts/gen_trace.py --no-sti --no-migration
```

```
# Generic STI only (no interval / mutex channels)
python3 scripts/gen_trace.py --no-intervals --no-mutex-sti
```

Options

Option	Default	Description
`-c` / `--cores`	4	Number of CPU cores
`-t` / `--tasks`	100	Number of worker tasks
`-e` / `--events`	8 000	Target non-comment event lines
`-o` / `--output`	auto	Output `.btf` file path
`--tick-hz`	1000	RTOS tick frequency in Hz (1000 → 1 ms per tick)
`--freq-hz`	100 000 000	CPU clock frequency in Hz (written to BTF header)
`--sti-interval-us`	30 000	Approximate µs between generic STI tag events
`--interval-ids`	3	Number of distinct interval IDs (`0` … `N-1`)
`--mutex-count`	2	Number of mutex objects (`create` / `take` / `give` STI)
`--idle-prob`	0.20	Probability [0–1] that a core picks its IDLE task
`--max-burst-ticks`	5	Maximum ticks a task runs before being preempted
`--no-sti`	off	Suppress all STI software-trace events
`--no-intervals`	off	Suppress `interval_start` / `interval_stop` STI

Option	Default	Description
`--no-mutex-sti`	off	Suppress mutex `create` / `take` / `give` STI
`--no-migration`	off	Pin each task to one core (disable migration)

When `--output` is omitted the file is named automatically, e.g. `freertos_8c_100t_1m_events.bt`

BTF format

Line structure

Every non-comment line is a comma-separated record with 7 or 8 fields:

`timestamp, source, src_inst, event_type, target, tgt_inst, event[, note]`

Field	Index	Type	Description
`timestamp`	0	integer	Absolute time in the unit declared by `#timeScale`
`source`	1	string	Entity that emits the event (core name or task label)
`src_inst`	2	integer	Source instance — always `0` in this implementation
`event_type`	3	string	`T`, `STI`, or `C` (see below)
`target`	4	string	Entity that receives the event (task label or STI channel)
`tgt_inst`	5	integer	Target instance — always `0` in this implementation

Field	Index	Type	Description
`event`	6	string	Event verb (`resume`, `preempt`, `trigger`, `set_frequency`, …)
`note`	7	string	Optional annotation (`task_create`, tick counter, mutex name, …). May be an empty string; a trailing comma is still present in that case.

Header comments

The file begins with #-prefixed metadata lines. The parser extracts key–value pairs of the form `#key value`:

```
#version 2.2.0
#creator synthetic_trace_gen
#creationDate 2024-01-01T00:00:00Z
#timeScale us
```

The value of `#timeScale` (`ns`, `us`, `ms`, …) determines the unit for every timestamp in the file.

Both parsers read `#version` and warn if it is present but not a **2.x** release (e.g. `#version 2.2.0` is supported). Meta keys must match `[\w.-]+` on Desktop and Web.

Core naming — `Core_N`

Cores are identified by the string `Core_` followed by a zero-based decimal integer:

```
Core_0  Core_1  Core_2  …  Core_N
```

`Core_0` is always the first core. The parser recognises a token as a core entity when it starts with `Core_`.

Task label naming — `[core_id/task_id]task_name`

Regular (worker) tasks carry a structured prefix that encodes the core they were created on and their unique task ID:

```
[core_id/task_id]task_name
```

Part	Example	Description
`core_id`	`0`	Zero-based index of the core that created this task
`task_id`	`9`	Unique integer task identifier assigned at task creation
`task_name`	`CAN_Rx`	Human-readable task name

> **Note:** In traces generated by `scripts/gen_trace.py`, worker task IDs start at 9 and the timer-service task ID equals `num_workers + 9`. Task IDs in real FreeRTOS ports depend on the kernel's internal handle allocation.

Examples:

The viewer accepts three task-name encodings (task id decimal or `0x` hex):

Form	Example	Display
`[core/task]name`	`[0/0001]MainCtrl`, `[2/0x9]CAN_Rx`	`MainCtrl[1]`, `CAN_Rx[0x9]`
`name[task]`	`MainCtrl[1]`, `Worker[0x8]`	same
`name(task)`	`Worker(0x8)`, `MyTask(42)`	`Worker[0x8]`, `MyTask[42]`

Legacy examples:

```
[0/9]CAN_Rx          # task CAN_Rx, created on Core_0, task ID 9
[2/17]Motor_L        # task Motor_L, created on Core_2, task ID 17
[0/42]Tmr Svc        # FreeRTOS timer-service task
```

The viewer displays these as `CAN_Rx[9]` and `Motor_L[17]` (task ID in brackets, core prefix hidden).

In **Task View** the viewer merges all instances of the same `task_id`/`task_name` pair across cores into one row, so a task that migrates between cores still appears as a single row.

Special tasks — no prefix IDLE and TICK tasks use a **bare name** with no `[core_id/task_id]` prefix:

Entity	Name pattern	Example	Notes
IDLE task	`IDLE` + core index	`IDLE0`, `IDLE1`, …	One per core, numbered from 0
Generic IDLE	`IDLE`	`IDLE`	Single-core systems
Tick ISR	`TICK`	`TICK`	System tick interrupt

IDLE tasks are always rendered in grey; each one gets a distinct shade. TICK tasks are rendered without a `[id]` suffix in labels.

event_type field

event_type	Description
T	Task context-switch event (task is resumed or preempted)
STI	Software Trace Item — application-level instrumentation marker
C	Core-level event (e.g. clock frequency change)

T events — task context switches

Each context switch produces two lines — one `preempt` and one `resume` — at the same timestamp. The **source** field rules are:

Switch type	`preempt` source	`resume` source
Timer-interrupt preemption	`Core_N` (the core that fired the interrupt)	Old task label (the task just preempted)
Voluntary yield (e.g. `vTaskDelay`)	Old task label	Old task label

The **target** is always the task label being directly affected: the task being stopped (`preempt`) or the task being started (`resume`).

event verb	Meaning
`resume`	Task begins executing on a core

event verb	Meaning
preempt	Task stops executing (preempted or blocked)

Examples:

```
# Timer interrupt preempts [0/9]CAN_Rx and resumes [0/12]Motor_L on Core_0
1000500, Core_0, 0, T, [0/9]CAN_Rx,  0, preempt,
1000500, [0/9]CAN_Rx, 0, T, [0/12]Motor_L, 0, resume,

# Task yields voluntarily (e.g. vTaskDelay)
2001000, [0/12]Motor_L, 0, T, [0/12]Motor_L, 0, preempt,
2001000, [0/12]Motor_L, 0, T, IDLE0, 0, resume,

# Task creation notification
  405, Core_0, 0, T, IDLE0, 0, preempt, task_create
  420, Core_0, 0, T, [0/9]CAN_Rx, 0, preempt, task_create

# TICK ISR fires (bare task name)
1000000, TICK, 0, T, TICK, 0, resume, tick_0
1000001, TICK, 0, T, TICK, 0, preempt,
```

STI events — software trace items

Source is the **core** (`Core_N`) that recorded the event. Target is the **STI channel name** (a free-form string that names the instrumentation point). The event verb is always `trigger`. The optional note field carries additional detail.

```
timestamp, Core_N, 0, STI, channel_name, 0, trigger[, note]
```

Examples:

```
3050000, Core_0, 0, STI, Mutex_Lock,    0, trigger, Mutex_Lock
3120000, Core_1, 0, STI, Queue_Send,    0, trigger, Queue_Send
3200000, Core_2, 0, STI, ISR_Enter,     0, trigger, ISR_Enter
3210000, Core_2, 0, STI, ISR_Exit,      0, trigger, ISR_Exit
```

Common STI channels generated by `scripts/gen_trace.py`:

Generic tags: `ISR_Enter`, `ISR_Exit`, `Sem_Post`, `Sem_Wait`, `Mutex_Lock`, `Mutex_Unlock`, `Queue_Send`, `Queue_Recv`, `Buf_Full`, `Buf_Empty`, `DMA_Done`, `DMA_Error`, `Overrun`, `Underrun`, `Checkpoint`, `Assert_OK`

Interval / mutex (firmware-style): `interval_start`, `interval_stop` (notes like `1` or `1 tid:9`); `mutex` with notes `create 0x80010100`, `take 0x80010100`, `give 0x80010100`

The viewer renders each distinct STI channel as a separate coloured row of diamond markers. Well-known notes (`take_mutex`, `give_mutex`, `create_mutex`, `trigger`) have fixed colours; others are assigned deterministically from a palette (same note → same colour every run).

FreeRTOS trace firmware (`gentrace` / `Demo`) Traces from `make run` and `tools/gentrace` use these STI channels (see Binary → BTF dump mapping for the full event list):

STI `target`	BTF `note` (examples)	Purpose
`interval_start`	`1 tid:7`	Interval region start (`id` + caller task id)
`interval_stop`	`1 tid:7`	Interval region end (pair with matching start + `tid`)
`tag0_event` … `tag7_event`	numeric payload	User tags (`traceTAG(t, v)`)

STI target	BTF note (examples)	Purpose
TICK	tick count	Scheduler tick (traceTASK_INCREMENT_TICK)
task	suspend Name[id], resume Name[id], delete …, set_priority Name[id] pri:N, priority_inherit …, priority_disinherit …, resume/isr	Task lifecycle / priority
queue / mutex / sem	create / send / recv / give / take / delete + 0x........	Queue and synchronisation objects

Interval example (from tracedata/example-4cores.btf.gz):

```
214276,Core_0,0,STI,interval_start,0,trigger,0 tid:1
217432,Core_1,0,STI,interval_stop,0,trigger,1 tid:7
```

The viewer pairs lines with the same note key: when tid is present, {id} tid:{task_id}; otherwise the full note string (legacy).

Task create on T rows uses note create pri:N (priority in param2), not task_create (synthetic scripts/gen_trace.py traces still use task_create).

C events — core events

Source and target are both the **core name**. Used for core-level notifications such as clock-frequency changes at startup.

```
timestamp, Core_N, 0, C, Core_N, 0, set_frequency, freq_hz
```

Example:

```
405, Core_0, 0, C, Core_0, 0, set_frequency, 200000000
410, Core_1, 0, C, Core_1, 0, set_frequency, 200000000
```

Complete annotated example

```
#version 2.2.0
#creator synthetic_trace_gen
#creationDate 2024-01-01T00:00:00Z
#timeScale us

# ── Startup: set clock frequency on every core ──────────────────────
405,  Core_0, 0, C,   Core_0,          0, set_frequency, 200000000
410,  Core_1, 0, C,   Core_1,          0, set_frequency, 200000000

# ── Create IDLE tasks ───────────────────────────────────────────────
415,  Core_0, 0, T,   IDLE0,           0, preempt, task_create
430,  Core_1, 0, T,   IDLE1,           0, preempt, task_create

# ── IDLE tasks start running ────────────────────────────────────────
480,  IDLE0,  0, T,   IDLE0,           0, resume,
490,  IDLE1,  0, T,   IDLE1,           0, resume,

# ── Create worker tasks (on Core_0) ─────────────────────────────────
510,  Core_0, 0, T,   [0/9]CAN_Rx,    0, preempt, task_create
528,  Core_0, 0, T,   [0/10]Motor_L,  0, preempt, task_create

# ── Normal context switches ─────────────────────────────────────────
1000000, TICK,            0, T, TICK,            0, resume,  tick_0
1000001, TICK,            0, T, TICK,            0, preempt,
1001500, Core_0,          0, T, IDLE0,           0, preempt,
1001500, [0/9]CAN_Rx,     0, T, [0/9]CAN_Rx,    0, resume,
1003000, [0/9]CAN_Rx,     0, T, [0/9]CAN_Rx,    0, preempt,
1003000, [0/9]CAN_Rx,     0, T, [0/10]Motor_L,  0, resume,

# ── STI software instrumentation ────────────────────────────────────
```

```
1050000, Core_0, 0, STI, Mutex_Lock,   0, trigger, Mutex_Lock
1120000, Core_1, 0, STI, Queue_Send,   0, trigger, Queue_Send
```

Contributors

Thanks to everyone who has contributed to this project!

Contributor	Contribution
DiogoRoseira	CPU Load Graph and Metrics distribution charts (scatter-plot + histogram popup)